Technology, Enforcement, and Quality in Agricultural Supply Chains: Experimental Evidence from Ugandan Dairy

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March 4, 2026

Abstract

Measurement technology could improve agricultural product quality through price incentives, through enforcement, or both. We develop a model nesting both channels and test its predictions in a field experiment providing digital milk analyzers to 120 milk collection centers serving 2,500 farmers in Uganda. Added water falls sharply and butterfat rises modestly, yet prices remain unchanged. An information intervention affects behavior only when paired with enforcement. The data consistently point to enforcement---raising the detection probability for substandard supply---rather than quality-differentiated pricing as the operative channel in this cooperative setting.

JEL: O12, O14, Q13, D82, L15

Keywords: quality enforcement, adulteration, measurement technology, dairy value chains, detection

1 Introduction

In many agricultural value chains in developing countries, quality is unobservable at the point of first sale. Milk, grain, tea, and other commodities are aggregated from multiple suppliers before reaching processors, and individual lots cannot be traced once pooled. When buyers cannot distinguish high- from low-quality supply, the classic adverse selection problem arises: producers who invest in quality are not rewarded, while those who adulterate their output face no penalty(Akerlof, 1970). The resulting equilibrium is characterized by low

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quality, widespread adulteration, and uniform pricing that reflects the pooled average rather than individual supply(Swinnen and Maertens, 2007; Barrett et al., 2022).

The conventional policy response follows a clear logic: make quality observable, enable quality-differentiated pricing, and let price premia incentivize quality investment(Shapiro, 1983; Leland, 1979). This"carrot" channel---measurement enables markets for quality---underpins much of the literature on value chain upgrading(Reardon and Barrett, 2000; Minten, Randrianarison, and Swinnen, 2009; Ponte and Gibbon, 2005). If buyers can verify compositional attributes and pay accordingly, producers have incentives to invest in better inputs and practices, and intermediaries have incentives to preserve quality during aggregation and transport(Hansman et al., 2020; Bold et al., 2022). Empirical evidence from dairy supply chains broadly supports this view: quality-differentiated pricing has been shown to improve milk composition in Indonesia(Treurniet, 2021) and Vietnam(Saenger et al., 2013), and group-level incentives can sustain quality improvements in Indian dairy cooperatives(Rao and Shenoy, 2023).

Yet in many settings, the institutional preconditions for quality-differentiated pricing are absent. Cooperatives maintain uniform prices to preserve solidarity among members. Concentrated processing sectors face little competitive pressure to differentiate on quality. Biweekly payment cycles lock in prices for fixed periods(Casaburi and Macchiavello, 2019). In such environments, even when measurement technology makes quality observable, the price channel may not operate. Does this mean that measurement technology cannot improve quality?

This paper provides evidence that it can---through a different mechanism. Drawing on the economics of enforcement and deterrence(Becker, 1968), we show that measurement technology improves quality through an*enforcement* channel: by raising the probability that substandard supply is detected and rejected. Even when prices remain flat and no quality premia emerge, the increased credibility of the accept/reject decision at the aggregation point raises the expected cost of delivering adulterated or low-quality milk. The"stick" of detection substitutes for the"carrot" of price premia. This mechanism parallels evidence from other contexts where increased monitoring reduces malfeasance(Olken, 2007; Duflo, Hanna, and Ryan, 2012) and where third-party verification disciplines behavior even without explicit incentive contracts(Duflo et al., 2013).

We study this mechanism in Uganda's dairy sector, where milk collection centers (MCCs) aggregate supply from thousands of smallholder farmers and sell to downstream processors. MCCs pay a flat price per liter regardless of compositional quality, and their existing testing capacity is limited to crude checks for freshness and gross adulteration(Van Campenhout, Minten, and Swinnen, 2021). Water addition to milk is endemic. We implement a multi-level randomized controlled trial across 120 MCCs and over 2,500 farmers in which we cross-randomize two interventions: (T1) digital milk analyzers installed at MCCs that enable precise, rapid compositional testing of every delivery; and (T2) an information treatment providing farmers with knowledge about compositional

quality parameters and practices to improve them.

We develop a model that nests both channels. A pricing parameter governs whether MCCs translate measurement capability into quality-differentiated payment: when it equals zero, the model reduces to pure enforcement; when positive, a price channel also operates. The two regimes generate sharply different predictions for prices, the magnitude of quality improvement, and whether farmer information works independently or only through enforcement. Under enforcement alone, the model predicts a large reduction in adulteration, a modest improvement in compositional quality bounded at the acceptance floor, no change in prices, and an information treatment that works only when paired with the analyzer.

The experimental results align closely with these predictions. MCCs adopted the analyzers and used them to screen incoming supply: testing of incoming and outgoing milk increased massively. Objective milk sample data show that added water fell sharply at treated MCCs, while butterfat increased modestly---direct evidence that the enforcement mechanism operated. Prices remained unchanged at every stage of the chain: MCC buying prices, MCC selling prices, farmer prices, and the prevalence of quality premia were all unaffected. The farmer-level information treatment had effects only when paired with the analyzer, confirming that enforcement and knowledge are complements: information about quality is actionable only when there is a credible threat of detection. Directly connected farmers responded more strongly than those selling through traders, consistent with the model's prediction that intermediation dilutes individual detection.

The paper makes three contributions. First, we provide a model that nests enforcement and price-incentive channels and use the experimental variation to identify which dominates. The existing literature on quality upgrading in agricultural value chains emphasizes the price channel---measurement enables price differentiation, which incentivizes quality investment(Treurniet, 2021; Saenger et al., 2013; Bold et al., 2022). We show that measurement can improve quality even when institutional constraints prevent quality-differentiated pricing from emerging. This distinction matters for policy: the enforcement channel operates within existing institutional structures, while the price channel requires complementary institutional changes in cooperative governance, competitive procurement, and payment systems.

Second, the theoretical framework parsimoniously explains the asymmetric pattern of results---large adulteration reduction, modest quality improvement, no price response---that would appear puzzling under the price channel hypothesis. The model clarifies why measurement technology is most effective on the extensive margin (deterring a costless but risky action) and less effective on the intensive margin (motivating costly investment with bounded returns), and why the absence of quality premia is not a disappointment but an expected feature of the enforcement-only equilibrium.

Third, the study contributes to a growing literature on the role of information and communication technologies in agricultural value chains(Van Campenhout, 2022; Jensen, 2007). While much of this literature focuses on how ICTs facilitate

price discovery or market access, we show that digital measurement tools can function as enforcement infrastructure---changing behavior through the threat of detection rather than through improved information transmission. This mechanism is relevant beyond dairy, in any commodity chain where adulteration is prevalent and aggregation points serve as natural enforcement nodes.

The rest of the paper is organized as follows. Section 2 describes Uganda's dairy value chain and the institutional context. Section 3 frames the competing channels through which measurement technology may affect quality and develops the research hypotheses. Section 4 presents the conceptual framework and derives testable predictions. Section 5 details the two intervention bundles and Section 6 outlines the experimental design. Section 7 discusses the estimation strategy. Section 8 describes the sample, data collection, and randomization. Section 9 examines baseline balance, and Section 10 documents attrition and compliance. Section 11 reports results at the MCC, farmer, and milk-sample levels, and discusses threats to validity and alternative explanations. Section 12 concludes.

2 Background: Dairy Supply Chains in Uganda

Uganda's dairy subsector has undergone rapid transformation over the past two decades. The privatization of the National Dairy Corporation and the enactment of the Dairy Industry Act of 1998 opened the sector to private investment and established the Dairy Development Authority (DDA) as the statutory regulator(Van Campenhout, Minten, and Swinnen, 2021). These reforms spurred an influx of foreign direct investment, particularly in and around Mbarara in southwestern Uganda---the country's primary milk shed---where a cluster of processing firms now converts raw milk into UHT milk, powdered milk, and infant formula for domestic and export markets. The sector is one of Uganda's main export earners.

A typical dairy value chain connects five actors. At the upstream end, smallholder *dairy farmers* produce milk and deliver it---either directly or through intermediaries---to *milk collection centers* (MCCs). *Transporters* collect milk from farms and deliver it to MCCs for a fee; *traders* purchase milk from farmers and resell it at MCCs for profit, often pooling supply from multiple producers. MCCs bulk and chill milk, initiating the cold chain, and are distributed across rural areas to facilitate access. Once chilled, *large traders* transport milk in tankers to *processors* concentrated in Mbarara and other towns.

Three features of this chain are critical to our study. First, *adulteration is endemic*. The most common form is the addition of water to milk(Rustagi and Kroell, 2022), which increases the volume a farmer can sell while degrading compositional quality. Because raw milk is opaque and its compositional attributes---butterfat, solids-not-fat (SNF), protein---are not visible to the naked eye, moderate water addition is difficult to detect without laboratory equipment. Adulteration is essentially costless to the farmer but imposes substantial costs downstream: it reduces processing yields, compromises food safety, and

degrades the quality of finished products.

Second, *pricing is flat*. Unlike in more developed dairy markets where processors pay per kilogram of milk solids, MCCs in Uganda typically pay a uniform price per liter regardless of butterfat, SNF, or protein content. This flat-price regime is sustained by several institutional features. Many MCCs are organized as cooperatives whose members resist price differentiation on grounds of solidarity and equity(Winfree and McCluskey, 2005; Bonroy et al., 2019). Prices are set on biweekly cycles that lock in a fixed rate for all deliveries within each period(Casaburi and Macchiavello, 2019). And the processing sector, while growing, is sufficiently concentrated that individual MCCs face limited competitive pressure to differentiate on quality. The result is a pricing structure that provides no financial reward for higher-quality supply and no financial penalty for lower-quality supply---as long as the milk is accepted.

Third, *MCCs are the natural enforcement node*, but they currently lack the capacity to screen quality effectively. The MCC is the first point in the chain where individual farmer deliveries can be evaluated before being pooled into a collective tank. Once milk enters the tank, traceability to individual suppliers is lost. The MCC therefore occupies a unique gatekeeper position: it is the last point at which an accept/reject decision on individual supply can be made. Yet most MCCs rely on only two crude diagnostic tests. The lactometer test uses a floating hydrometer to measure density as a proxy for water addition, but readings are sensitive to temperature and cannot distinguish intentional adulteration from natural variation in composition. The alcohol test assesses protein stability as a proxy for freshness---milk that curdles upon contact with ethanol is rejected as sour. Neither test provides information on compositional quality. The enforcement threshold that MCCs can credibly impose is therefore low: only gross adulteration or spoilage is detected, while moderate water addition and compositional variation go unscreened.

These three features---prevalent adulteration, flat pricing, and weak screening at the aggregation point---define the setting into which we introduce measurement technology. The question is whether equipping MCCs with precise, rapid milk analyzers changes farmer behavior, and if so, through what channel.

3 Research Questions and Hypotheses

The central research question of this paper is: through what channel does measurement technology affect product quality? The literature offers two competing hypotheses.

3.1 Two competing channels

The *price channel* is the conventional account. When quality becomes observable, buyers can condition payments on measured attributes, and quality premia emerge. These premia reward quality-enhancing investments and penalize low-quality supply, aligning incentives throughout the chain(Swinnen and Maertens,

2007; Reardon and Barrett, 2000). Under this channel, the causal chain runs: measurement→ price differentiation→ quality improvement. The price channel requires not only measurement capacity but also the institutional ability and competitive incentive to implement quality-differentiated payment schedules.

The *enforcement channel* offers an alternative mechanism. Even when prices remain flat, measurement technology improves the precision of the accept/reject decision at the aggregation point. By raising the probability that substandard milk is detected---much as third-party verification constrains self-reported behavior in other settings(Kleven et al., 2011)---the analyzer increases the expected cost of delivering adulterated or low-quality supply. Under this channel, the causal chain is: measurement→ higher detection probability→ increased rejection risk→ quality improvement. The enforcement channel can operate within existing institutional structures---it requires only that the MCC has both the capacity and the willingness to reject substandard supply.

These two channels generate sharply different predictions. If the price channel dominates, we should observe quality premia emerging after analyzer installation, price increases for higher-quality supply, and quality improvement driven primarily by the intensive margin (investments in feed, breed, and management practices that raise compositional quality). If the enforcement channel dominates, we should observe no change in prices, no emergence of quality premia, and quality improvement driven primarily by the extensive margin (cessation of adulteration) with modest intensive-margin responses. Section4 develops a model that nests both channels and derives testable predictions that distinguish between them.

3.2 Hypotheses

Our experimental design tests four main hypotheses. These correspond to the predictions of the enforcement channel but, as Section4 formalizes, the model's two-channel structure generates contrasting predictions that allow the data to reveal which channel dominates.

The first hypothesis concerns MCC-level uptake and outcomes. Installing milk analyzers at MCCs (T1) enables precise, rapid screening of incoming supply. Under the enforcement channel, MCCs should adopt the technology and use it to enforce quality thresholds more effectively. This should raise the average quality of aggregated milk, primarily through rejection of adulterated supply, even without changes in pricing.

The second hypothesis concerns farmer-level responses to midstream enforcement. When MCCs acquire the ability to detect adulteration and compositional deficiencies, the expected cost of delivering substandard milk rises for supplying farmers. Under the enforcement channel, farmers should respond primarily on the extensive margin---reducing or eliminating water addition---because the private gain from adulteration is small relative to the revenue lost upon rejection. Intensive-margin responses (investing in feed and handling to raise compositional quality) should be modest, because the flat-price regime offers no incremental reward for quality above the acceptance threshold.

The third hypothesis concerns the farmer-level information intervention (T2). Farmers in Uganda often associate milk quality with hygiene rather than compositional parameters, and may lack knowledge of the feeding practices that affect butterfat and SNF.[1] The information treatment provides this knowledge, effectively increasing the productivity of quality effort. However, under the enforcement model, information alone should have limited effects: without a credible threat of detection and rejection, there is little return to improving quality. The information treatment should therefore be most effective when combined with the analyzer (T1×T2), reflecting a complementarity between enforcement and knowledge.

The fourth hypothesis concerns the role of intermediation. Farmers who sell through traders rather than delivering directly to MCCs face a weaker enforcement link: traders pool milk from multiple farmers before delivery, diluting the traceability of individual quality to its source. The enforcement channel therefore predicts that directly connected farmers should respond more strongly to analyzer installation than trader-connected farmers(Rao and Shenoy, 2023; Mitra et al., 2018).

4 Conceptual Framework and Testable Predictions

This section develops a model of farmer behavior that nests two channels through which measurement technology may improve quality. The model introduces a parameter$\alpha \in [0, 1]$ that governs whether MCCs translate measurement capability into quality-differentiated pricing. When$\alpha = 0$, the MCC pays a flat price for all accepted milk and the model reduces to pure enforcement: the analyzer operates solely by raising the probability that substandard milk is detected and rejected. When$\alpha > 0$, a price channel also activates: the MCC pays a premium for quality above the floor, creating a continuous return to quality investment. The model abstracts from the multi-component nature of the intervention bundle (Section 5) to focus on two parameters---the detection probabilityδ and the institutional pricing parameterα---through which the analyzer operates. As discussed in Section 5, the ancillary components (app, tablet, DDA visits) support the functioning of the analyzer but do not independently generate quality signals. The two regimes generate sharply different predictions for prices, the magnitude of quality improvement, and the interaction between enforcement and farmer information, allowing the data to reveal which channel dominates.

[1] Agricultural extension services tend to prioritize food safety over compositional quality. As a result, farmers are typically trained on sanitary standards but receive little guidance on how feeding practices, forage selection, or supplementation affect butterfat and SNF.

4.1 Environment and Timing

There are multiple MCCs indexed by m, and within each MCC catchment, multiple farmers indexed by i deliver milk to the MCC. Each farmer supplies a fixed quantity of raw milk y_i per period, determined by herd size and not adjustable in the short run. The MCC pays a flat per-unit price p_0 for accepted milk. Milk that is rejected earns a salvage value $s \geq 0$, with $s < p_0$.[2]

More generally, the MCC may also pay a quality premium for milk above the floor. We model the price received by farmer i as:

$$p_i = p_0 + \alpha\delta\beta\left(q_i - \underline{q}\right)^+ \tag{1}$$

where $\alpha \in [0, 1]$ captures the MCC's institutional willingness or capacity to implement quality-differentiated pricing, δ is the measurement precision (the MCC can only price on quality it can observe), $\beta > 0$ is the price-quality gradient when the price channel is active, and $\left(q_i - \underline{q}\right)^+ \equiv \max\left(q_i - \underline{q}, 0\right)$ ensures that the premium applies only to quality above the floor. When $\alpha = 0$---the flat-pricing regime---the price reduces to p_0 for all accepted milk, and the model operates through enforcement alone. When $\alpha > 0$, the price channel is active: the MCC translates its measurement capability into quality premia. Note that the price channel requires *both* measurement precision ($\delta > 0$) and institutional willingness ($\alpha > 0$).

A critical feature of the setting is that the MCC applies a minimum quality threshold $\underline{q}$: milk is accepted if observed quality meets or exceeds this floor, and rejected otherwise. However, the MCC's ability to enforce this threshold depends on the precision of its quality measurement technology.

4.2 Farmer Actions: Quality Effort and Adulteration

Each farmer i makes two choices that jointly determine the quality of delivered milk. First, the farmer chooses a level of quality effort $e_i \geq 0$ that reflects investments in feeding, hygiene, and handling practices. Second, the farmer chooses an adulteration level $w_i \geq 0$, representing the amount of water (or other adulterants) added to the milk. The quality of delivered milk is:

$$q_i = \theta_i e_i - \gamma w_i + \varepsilon_i \tag{2}$$

where $\theta_i > 0$ captures how effectively effort translates into compositional quality (e.g., butterfat), $\gamma > 0$ is the rate at which adulteration degrades quality, and ε_i is a mean-zero stochastic shock reflecting natural variation.

Quality effort is costly. We model the cost of effort as a convex function:

$$c(e_i) = \frac{k}{2}e_i^2 \tag{3}$$

[2] In practice, rejected milk can sometimes be sold on the informal market at a steep discount, or it may be a total loss. The key assumption is that rejection is costly: $s < p_0$.

with $k > 0$. Adulteration, by contrast, is essentially costless to the farmer---adding water requires negligible input cost---but generates a private benefit $b > 0$ per unit of adulterant added, reflecting the volume gain from stretching the milk supply:

$$B(w_i) = bw_i \tag{4}$$

4.3 Detection and the Probability of Rejection

The MCC maintains a minimum quality threshold $\underline{q}$ and rejects milk that falls below it. However, the MCC can only enforce this threshold to the extent that it can actually measure quality. Without precise equipment, low-quality or adulterated milk may go undetected and be accepted. We model this as a detection probability $\delta \in [0, 1]$: the probability that the MCC accurately observes whether $q_i \geq \underline{q}$.

The probability that farmer i's milk is rejected is:

$$\rho_i(e_i, w_i; \delta) = \delta \cdot \Pr\left(q_i < \underline{q}\right) \tag{5}$$

When δ is low (poor measurement technology), even milk that truly falls below the threshold is likely accepted because the MCC cannot identify it. When δ is high (precise analyzer available), substandard milk is detected and rejected with high probability.

For a farmer whose true quality exceeds the floor $\left(\theta_i e_i - \gamma w_i > \underline{q}\right)$, the rejection probability is approximately zero regardless of δ.[3] For a farmer whose true quality falls below the floor, the rejection probability is δ. The key comparative static is:

$$\frac{\partial \rho_i}{\partial \delta} \geq 0, \quad \text{with strict inequality when } q_i < \underline{q} \tag{6}$$

Better measurement technology increases the rejection probability only for farmers currently delivering substandard milk.

4.4 Farmer Optimization

The farmer's expected payoff from delivering milk includes both the acceptance/rejection lottery and, when $\alpha > 0$, a quality premium on accepted milk above the floor. Using the price schedule from Equation 1, the payoff can be written as:

$$V_i(e_i, w_i) = [1 - \rho_i]\,(p_0 - s)\,y_i + sy_i + \alpha\delta\beta\mathbb{E}\left[\left(q_i - \underline{q}\right)^+\right] y_i + bw_i - \frac{k}{2}e_i^2 \tag{7}$$

The first term is the stake at risk from rejection, the second is baseline salvage revenue, the third is expected quality premium revenue (zero when $\alpha = 0$), the fourth is the private benefit of adulteration, and the last is the cost of quality effort. Denote the stake at risk as $L \equiv (p_0 - s)\,y_i > 0$.

[3] We abstract from the noise term ε_i in the discussion below for clarity. Formally, the rejection probability also depends on the variance of ε_i, but this does not change the qualitative comparative statics.

The farmer chooses e_i and w_i to maximize V_i. Consider the two margins in turn.

4.4.1 Adulteration decision (extensive margin)

The marginal benefit of adding water is the volume gain b. The marginal cost includes both the increase in rejection probability times the stake and, when $\alpha > 0$, the reduction in the quality premium:

$$\frac{\partial V_i}{\partial w_i} = b - \gamma \left[\delta f(\underline{q}) \cdot L + \alpha\delta\beta y_i\right] \tag{8}$$

The first term in brackets is the enforcement cost (rejection risk); the second is the price-channel cost (lost quality premium). Since adulteration degrades quality at rate γ, and $f(\cdot)$ is the density of q_i evaluated near the threshold, the farmer stops adulterating when detection is sufficiently high:

$$w_i^*(\delta) = \begin{cases} w^+ > 0 & \text{if } \delta < \hat{\delta}_w(\alpha) \equiv \frac{b}{\gamma\left[f(\underline{q})\cdot L + \alpha\beta y_i\right]} \\ 0 & \text{if } \delta \geq \hat{\delta}_w(\alpha) \end{cases} \tag{9}$$

This is an extensive-margin response: at the critical detection threshold $\hat{\delta}_w(\alpha)$, the farmer discontinuously stops adding water. When $\alpha = 0$, the threshold depends only on the enforcement stake; when $\alpha > 0$, the threshold is strictly lower because adulteration also forfeits quality premia. In both regimes, the private gain from adding water is small relative to the penalties, so even a moderate increase in detection probability can eliminate adulteration entirely.

4.4.2 Quality effort decision (intensive margin)

The first-order condition for quality effort equates the marginal return---from both reduced rejection risk and, when $\alpha > 0$, the increased quality premium---to the marginal cost:

$$\delta\theta_i \left[f(\underline{q}) \cdot L + \alpha\beta y_i\right] = ke_i \tag{10}$$

At an interior solution:

$$e_i^* = \frac{\delta\theta_i}{k} \left[f(\underline{q}) \cdot L + \alpha\beta y_i\right] \tag{11}$$

The enforcement return $f(\underline{q}) \cdot L$ is now augmented by a price-channel return $\alpha\beta y_i$. Under $\alpha = 0$, optimal effort is proportional to δ but bounded by the density $f(\underline{q})$: once the farmer's quality comfortably exceeds the floor, the marginal rejection-reducing benefit of further effort is negligible, so effort plateaus. The enforcement channel motivates farmers to clear the floor, not to invest heavily in quality above it. Under $\alpha > 0$, the return to effort continues above the floor because quality earns a price premium. Effort is higher for any given δ, and the diminishing-returns feature is weaker because the price-channel return $\alpha\beta y_i$ does not depend on the floor density.

4.5 Comparative Statics: Effect of the Milk Analyzer

The milk analyzer increasesδ from a low levelδ_0 (traditional testing, e.g., lactometer and alcohol tests) to a high level$\delta_1 > \delta_0$ (precise compositional analysis). The comparative statics differ across the two regimes. For each prediction below, we state what the model implies under$\alpha = 0$ (enforcement only) and under$\alpha > 0$ (price channel active), and note whether the prediction distinguishes between them.

4.5.1 Prediction 1: Reduction in adulteration (not distinguishing)

From Equation9, both regimes predict that adulteration falls whenδ increases. Under$\alpha = 0$, adulteration drops discontinuously to zero onceδ exceeds$\hat{\delta}_w(0) = b/[\gamma f(\underline{q})L]$. Under$\alpha > 0$, the threshold is strictly lower---$\hat{\delta}_w(\alpha) < \hat{\delta}_w(0)$---because adulteration also forfeits quality premia. In both cases, the private gain from adding water is small relative to the penalties, so moderate improvements in detection deter adulteration for most farmers. This prediction does not distinguish between regimes:

$$\frac{\partial w_i^*}{\partial \delta} \leq 0 \quad \text{under both } \alpha = 0 \text{ and } \alpha > 0 \tag{12}$$

4.5.2 Prediction 2: Compositional quality improvement (distinguishing)

From Equation11, quality effort is increasing inδ under both regimes, but the magnitude differs. Under$\alpha = 0$, the return to effort comes only from reducing rejection probability near the floor: once quality comfortably exceeds$\underline{q}$,$f(\underline{q})$ becomes small and effort plateaus. The predicted improvement is therefore*modest*, concentrated among farmers near the threshold. Under$\alpha > 0$, the additional return$\alpha\beta y_i$ persists above the floor, so effort is*larger* and quality improvement is more substantial:

$$e_i^*(\alpha > 0) > e_i^*(\alpha = 0) \quad \text{for all } \delta > 0 \tag{13}$$

4.5.3 Prediction 3: Prices (sharply distinguishing)

From Equation1, the price prediction provides the sharpest test between regimes. Under$\alpha = 0$, the MCC pays a flat price for all accepted milk, and increased measurement precision has no effect on prices:

$$\alpha = 0: \quad \frac{\partial p_i}{\partial \delta} = 0 \tag{14}$$

Under$\alpha > 0$, improved measurement enables quality-differentiated pricing and quality premia emerge:

$$\alpha > 0: \quad \frac{\partial p_i}{\partial \delta} = \alpha\beta\left(q_i - \underline{q}\right)^+ > 0 \quad \text{for } q_i > \underline{q} \tag{15}$$

Flat prices after analyzer installation indicate$\alpha \approx 0$; the emergence of quality premia would indicate$\alpha > 0$.

4.5.4 Prediction 4: Asymmetric responses (distinguishing)

Both regimes predict that adulteration reduction exceeds quality improvement, because stopping a costless action is easier than investing in costly effort. However, the degree of asymmetry differs. Under$\alpha = 0$, the asymmetry is large because quality effort is bounded by the floor density:

$$\alpha = 0: \quad \left|\frac{\partial w_i^*}{\partial \delta}\right| \gg \left|\frac{\partial e_i^*}{\partial \delta}\right| \tag{16}$$

Under$\alpha > 0$, the asymmetry is smaller because quality effort has a continuous price return above the floor. A very large asymmetry---large drop in added water, modest increase in fat, no change in SNF or protein---favors$\alpha \approx 0$.

4.5.5 Prediction 5: MCC uptake and screening (not distinguishing)

From the MCC's perspective, the analyzer provides a better signal of incoming quality at low marginal cost per test. Under both regimes, a more precise signal enables more effective enforcement (and, under$\alpha > 0$, more accurate pricing), which raises average quality$\bar{q}_m$ in the aggregated tank. MCCs therefore have a strong incentive to adopt and use the analyzer regardless of the pricing regime:

$$\bar{q}_m(\delta_1) > \bar{q}_m(\delta_0) \quad \text{under both } \alpha = 0 \text{ and } \alpha > 0 \tag{17}$$

4.5.6 Prediction 6: Interaction between enforcement and farmer information (sharply distinguishing)

The video intervention increasesθ_i (the productivity of effort in producing quality). This prediction provides the key diagnostic between regimes. From Equation11, the main effect of training on effort is:

$$\frac{\partial e_i^*}{\partial \theta_i} = \frac{\delta}{k}\left[f(\underline{q}) \cdot L + \alpha\beta y_i\right] \tag{18}$$

Under$\alpha = 0$: training has*no main effect* whenδ is low, because without enforcement there is no return to effort. A significant positive interaction$T1 \times T2$ should be observed---training works only when paired with the analyzer. Under$\alpha > 0$: training has a*positive main effect* even without the analyzer, because quality effort earns a price premium whenever$\alpha\delta > 0$. The interaction is smaller and incremental rather than the dominant effect. In both regimes the cross-partial is positive:

$$\frac{\partial^2 e_i^*}{\partial \delta\, \partial \theta_i} > 0 \tag{19}$$

The diagnostic is whether the interaction dominates ($\alpha \approx 0$: training works only through enforcement) or the main effect dominates ($\alpha > 0$: training works through price incentives).

4.5.7 Prediction 7: Intermediation (not distinguishing)

Farmers who sell through traders face a longer and less transparent transaction chain. If traders pool milk from multiple farmers before delivery, the effective detection probability is lower:$\delta_T < \delta_C$. Under both regimes, this implies that directly connected farmers respond more strongly to the analyzer. The intermediation prediction does not distinguish between channels:

$$|\Delta w_i^*(\delta|\text{direct})| > |\Delta w_i^*(\delta|\text{trader})| \quad \text{under both } \alpha = 0 \text{ and } \alpha > 0$$

Summary. Table 1 summarizes the predictions under both regimes. Three predictions---compositional quality, prices, and the training interaction---sharply distinguish between$\alpha = 0$ and$\alpha > 0$. The remaining predictions hold in both regimes.

Table 1: Model predictions under enforcement-only ($\alpha = 0$) and price-channel ($\alpha > 0$) regimes

Prediction	$\alpha = 0$ **(Enforcement)**	$\alpha > 0$ **(Price channel)**	**Distinguishing?**
1. Adulteration	Large reduction	Large reduction	No
2. Quality	Modest, bounded at floor	Larger, continuous	Yes
3. Prices	No change	Quality premia emerge	Yes (sharp)
4. Asymmetry	Large	Smaller	Yes
5. MCC uptake	Strong adoption	Strong adoption	No
6. Training×Analyzer	Interaction dominates; no main effect	Main effect positive; interaction incremental	Yes (sharp)
7. Intermediation	Weaker for traders	Weaker for traders	No

4.6 Discussion

The model nests both channels through a single parameterα. The empirical pattern maps onto$\alpha \approx 0$: prices remain flat despite improved measurement, quality gains are modest and bounded at the acceptance floor, and the information treatment affects behavior only when paired with enforcement. Together, these results indicate that institutional constraints---cooperative pricing norms, limited competition among MCCs for suppliers, and organizational inertia---prevent the price channel from activating. Measurement technology is necessary but not sufficient for quality-differentiated markets; translating measurement capability into price incentives requires complementary institutional changes in cooperative governance, competitive procurement, or processor-imposed quality-price schedules.

The enforcement channel alone is sufficient to account for the central empirical findings: strong MCC uptake of the analyzers, a large decline in added

water, a modest increase in fat content, no change in prices, and no emergence of quality premia. The model clarifies why: in the$\alpha = 0$ regime, the analyzer raises the expected cost of delivering substandard milk, which deters adulteration (cheap to stop) more than it encourages quality investment (expensive to pursue and bounded in its returns under flat pricing).

5 Innovation Bundles for Addressing Milk Quality Challenges

In response to the challenge of unobservable milk quality in the Ugandan dairy value chain, we designed a first socio-technical innovation bundle targeting Milk Collection Centers (MCCs). This bundle aims to enhance transparency, improve record-keeping, and empower both MCC managers and farmers by making milk quality measurable and visible. The bundle, developed after extensive consultations with stakeholders and implemented together with the DDA, consists of three key components:

1. Milk Analyzer: A central component of the innovation bundle is the installation of milk analyzers at MCCs. These machines assess milk quality based on a set of compositional parameters, such as butterfat content and solids-not-fat (SNF). Another important parameter is the amount of added water, providing a more accurate assessment of what MCC managers usually test using gravity based methods. The testing process is non-destructive and takes less than one minute per sample, enabling rapid and accurate quality evaluation of each delivery.[4] By providing immediate feedback, the milk analyzers help ensure that quality standards are met and maintained. More information on the milk analyzer can be obtained from the manufacturer's website. Figure 1 shows a milk analyzer during piloting.

The milk analyzers were delivered with clear Standard Operating Procedures. Two separate trainings were organized. One training targeted MCC owners, where the focus was mostly on generating buy-in by pointing out the benefits of measuring and tracking milk quality. A second training was geared toward MCC managers and focused more on use and maintenance of the milk analyzers.

We collaborated with the DDA to set up a system to monitor the milk analyzers and its use. In particular, DDA technicians visited treatment MCCs at set periods. We also provided various ways in which MCC managers could request assistance if needed, including a WhatsApp group and a telephone hotline. Over the course of the project, we ensured that equipment was adequately calibrated and that MCC managers had access to cleaning reagents. If milk analyzers malfunctioned, we fixed or replaced the machines.

2. ICT-Mediated Record-Keeping System: We developed an Android application for MCC managers to facilitate digital record-keeping. The app was

[4]Generally milk was tested per (25 or 50 liter) milk can. Sometimes, if multiple cans of the same farmer were delivered, a composite sample of the different cans were tested.

Figure 1: Milk analyzer results during testing showing results for an exceptionally high quality milk sample

designed to replace the paper notebooks that MCC managers generally use for keeping track milk deliveries and payments. In addition to recording quantities and prices, the app allows MCC managers to store and monitor quality parameters obtained from the milk analyzer. The app can provide MCC managers with simple reports, such as the average butter fat (weighted by quantities supplied) over a certain period (today, yesterday, last week, last two weeks, and custom data range). Reports by farmer are also possible, such that MCC managers can determine the total sum to be paid to a farmer for milk delivered over a particular time frame, such as in the last 14 days.[5] In addition to the app, we also developed different online portals that can be used to obtain data for different stakeholders. For instance, one portal targeted MCC owners, such that they can monitor key parameters in the MCCs they own. Another portal aggregated information from all MCCs, enabling government officials such as the DDA to monitor quality parameters, prices and quantities in real time.[6] This digital system enhances efficiency and transparency, providing both MCC managers and farmers with reliable records that integrate milk quality metrics. The android app can be downloaded from google play store; A screenshot of the application can be found in Figure 2.

Milk collection centers were provided with Samsung Galaxy Tab A7 Android tablet computers on which the application was pre-installed. Each tablet contained a SIM card with a prepaid data bundle to enable cloud-based synchronization of records. The application was developed following offline-first design principles, ensuring that all core functionalities remained accessible without an active internet connection and that data were uploaded automatically once connectivity was restored. Data bundles were topped up by us on a monthly basis throughout the course of the project.

3. Increasing Farmer Awareness of Midstream Testing Capacity: To mitigate potential power imbalances that could arise once MCCs adopt milk analyzers and to ensure farmers are equally informed, we implemented a farmer-facing information campaign using posters displayed at MCCs. Designed by a local artist, the posters publicized the new testing capacity and encouraged farmers to request free quality tests for their milk. By making this service visible and widely known, the campaign aimed to equip farmers with accurate information about their product quality and strengthen trust and cooperation between farmers and MCC managers.

Together, these three components form an integrated strategy to make milk quality observable and actionable, improving information flows and enforcement capacity throughout the dairy value chain. This bundle constitutes the first treatment arm in the field experiment and is referred to as T1 (or sometimes also as the*analyzer* treatment).

We emphasize that while the T1 intervention is a package, the milk analyzer is the binding constraint. The record-keeping application and tablet serve as complementary infrastructure that enables managers to store and act on an-

[5]Farmers are typically paid on the 1st and the 15th of the month.

[6]The portal can be found here.

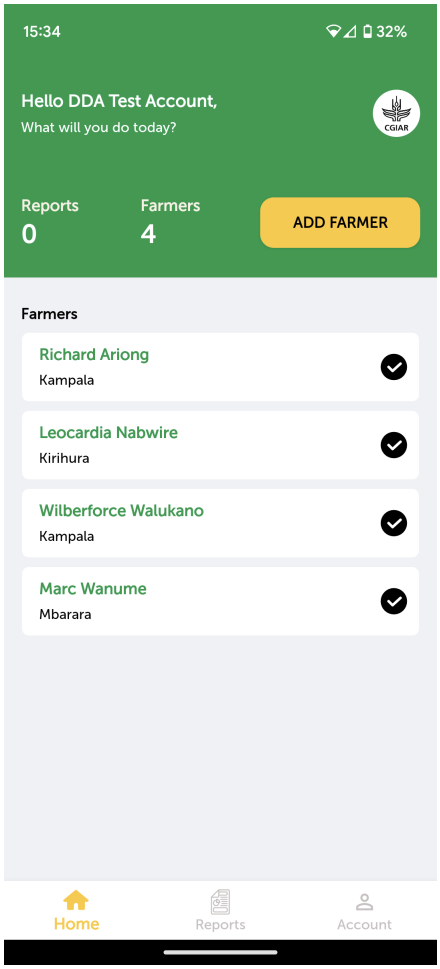

Figure 2: Dairy record keeping app

alyzer readings---they generate no quality signal on their own. Similarly, the farmer awareness posters inform farmers that testing occurs, but without the analyzer there would be nothing to test with. The DDA monitoring visits and calibration support ensure the equipment functions as intended, analogous to maintenance of any capital good. In other words, the supporting components make the analyzer effective but do not constitute independent treatment channels. We cannot experimentally isolate the analyzer from its support infrastructure, but we note that none of the ancillary components alone would increase detection probability---only the analyzer generates the objective quality measurement that makes credible enforcement possible. Our theoretical model therefore focuses on the detection probability parameterδ as the mechanism through which the package operates, while acknowledging that the empirical estimates capture the joint effect of the full bundle.

To address the mismatch between how farmers and processors perceive milk quality, we designed an intervention to improve farmers' understanding of compositional quality and its role in the dairy value chain. The treatment combines targeted information, practical guidance, and modest material support. It consists of three components:

1. Educational Video: A short, engaging video was developed to explain the concept of compositional milk quality (parameters such as butterfat and protein content) and why it matters for both farmers and processors. The video also highlights practical management practices and inputs that farmers can adopt to improve milk quality, mostly focusing on feeding strategies.[7] The video is designed to be accessible and appealing, ensuring key messages resonate with the target audience. The video was shown to treatment farmers twice: once during baseline data collection in December 2022 and once at the time of installation of the milk analyzers in October 2023. The use of videos such as the one we use has been found to increase technology adoption in different settings (Spielman et al., 2021).

2. Cartoon Handout: To reinforce the information provided in the video, we created a handout in the form of a cartoon summary. The cartoons provide a visual, easy-to-understand recap of the key points from the video, serving as a quick reference for farmers after watching the video. This format ensures the information remains accessible even to farmers with limited literacy.

3. Improved Pasture Seeds: To make the knowledge actionable, each participating farmer received a free bag of improved pasture seeds (1 kg of *Chloris Gayana*, also known as Rhodes grass). By planting these seeds, farmers can enhance the nutritional value of their pastures, a critical factor in increasing the compositional quality of milk. This practical input complements the edu-

[7] To determine the content of the video, we first identified the top five practices and inputs that are known to raise butter fat and SNF in milk. This was done through consultations of experts. We found the top 5 practices and inputs were: selection of breed and genetic potential, selection of grasses for high-quality forage, best practice in silage and hay making, correct mixing and dosage of feed, and feed supplements like Methionine and Lysine. As selection of breed and genetic potential is unlikely to change sufficiently fast give the length of our research project, we decided to focus on feeding practices.

cational components, enabling farmers to directly apply the recommendations provided.

This socio-technical innovation bundle, referred to as T2 (or sometimes also as the *video* treatment), seeks to narrow perception gaps around milk quality and strengthen farmers' capacity to supply higher compositional quality on a sustained basis.

6 Experiment

We use a field experiment to test the predictions in Section 4.5. In particular we use a split-plot design with interventions at two levels of the value chain. The design is illustrated in Figure 3.

The experimental design mirrors the structure of the dairy value chain and involves randomization at two levels. First, at the MCC level, we assign half of the MCCs to receive the first socio-technical innovation bundle (T1), with the remaining MCCs serving as controls. Second, within each MCC—regardless of its assignment to T1—we randomize the second innovation bundle (T2) at the farmer level. For each MCC, half of its supplying farmers are assigned to T2 and the remainder to the farmer-level control. This yields a nested, multi-level design in which T1 varies across MCCs and T2 varies within MCCs. To examine heterogeneous effects, we further stratify the farmer-level randomization by the nature of the farmer's linkage to the MCC (direct delivery versus delivery through a trader).

This multi-level design enables us to estimate impacts at different nodes of the value chain. At the MCC level, we can assess the effect of T1 on outcomes at the MCC level. Because T1 is assigned downstream, we can also evaluate its upstream effects on farmers linked to those MCCs. In contrast, the farmer-level treatment T2 varies only within MCCs, so its effects can be identified exclusively from farmer-level outcomes.

In summary, and in reference to the estimating equations introduced in the next section (Section7), the hypotheses developed in Section3, and the predictions derived in Subsection4.5, the four main hypotheses that we test with this design are:

- Hypothesis 1: installing the milk analyzer at the MCC level (T1) increases MCC uptake of quality screening and improves MCC-level outcomes($\beta_{H1} > 0$).
- Hypothesis 2: the enforcement capacity created by the analyzer (T1) reduces adulteration and improves quality outcomes for farmers in the catchment areas of treated MCCs($\beta_{H2} > 0$).
- Hypothesis 3: the information intervention (T2) increases farmer quality effort, but primarily when combined with the enforcement technology (T1)($\beta_{H3} > 0$).

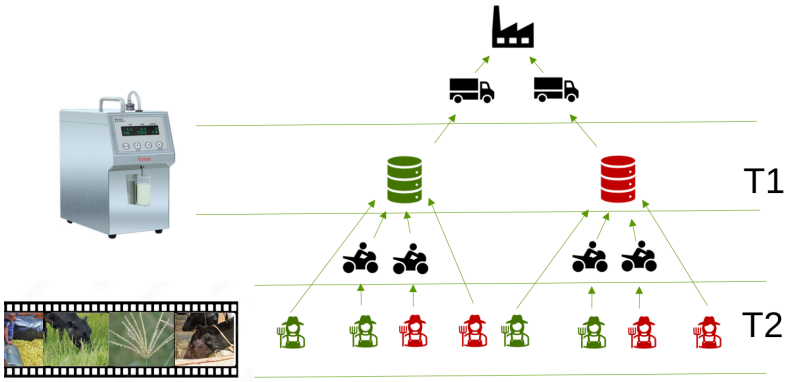

Figure 3: Design

- Hypothesis 4: the combined intervention (T1×T2) generates complementary effects on farmer outcomes through the interaction of enforcement and knowledge($\beta_{H4} > 0$).

Additional hypotheses test whether treatment effects differ by the farmer's mode of connection to the MCC, as predicted by the enforcement model's intermediation prediction:

- Hypothesis 5: the treatment effect of the analyzer (T1) is larger for farmers directly connected to the MCC than for those selling through traders($\beta_{H2C} \neq \beta_{H2T}$).
- Hypothesis 6: the treatment effect of the information intervention (T2) differs by mode of connection($\beta_{H3C} \neq \beta_{H3T}$).
- Hypothesis 7: the treatment effect of the combined intervention (T1×T2) differs by mode of connection($\beta_{H4C} \neq \beta_{H4T}$).

7 Estimation and Inference

To assess impact of the treatments, we estimate various specifications using Ordinary Least Squares. A first specification uses data at the MCC level. Denote milk collection centers by m, running from 1 to M. $T1_m$ is a treatment indicator at the MCC level that is one if the MCC was allocated to T1. y_m is the outcome of interest at the level of the milk collection center we want to estimate the treatment effect for and ε_m is an error term. In all equations we also control for baseline outcome if information was collected at that time $\left(y_m^b\right)$:

$$y_m = \alpha + \beta_{H1}.T1_m + \beta_b.y_m^b + \varepsilon_m \tag{20}$$

The parameter of interest in this equation is β_{H1}, which tests Hypothesis 1 in Section 6.

All other equations use farmer level data. We start by estimating the main treatment effect for T1 on farmers that are associated to that MCC. One way to do this is to estimate a fully interacted model such as:

$$y_{i,m} = \alpha + \beta^*_{H2}.T1_{i,m} + \beta^*_{H3}.T2_{i,m} + \beta^*_{H4}.T1_{i,m}.T2_{i,m} + \beta_b.y^b_{i,m} + \varepsilon_{i,m} \quad (21)$$

Here, $T2_i$ is a treatment indicator at the farmer level that is one if the farmer was allocated to T2, with i indicating the farmer running from 1 to I, $y_{i,m}$ is the outcome of interest at the level of the individual farmer living in the catchment area of milk collection center m and $\varepsilon_{i,m}$ is an error term (which may be correlated within catchment area). As above, $T1_{i,m}$ is a treatment indicator that is one if the MCC (in whose catchment area farmer i resides) was allocated to T1, and zero otherwise. We also control for the lagged baseline outcome $y^b_{i,m}$. The parameter of interest in this equation is β^*_{H2}, which tests the impact of T1 om farmers, corresponding to Hypothesis 2 in Section 6.

Equation 21 can also be used to test 2 other hypotheses. First, β^*_{H3}, provides an estimate of the farmer level impact of T2, which tests Hypothesis 3 from Section 6. Second, in a fully saturated model such as the one in Equation 21, β^*_{H4} also allows us to tests for the interaction effect, corresponding to Hypothesis 4 in Section 6.

While our power calculations relied on models that include the full set of interactions, an important feature of split-plot (factorial) designs is that statistical power to estimate the effect of one treatment can be increased by pooling observations across the orthogonal treatment (List, Sadoff, and Wagner, 2011). In our case, this means that the effect of the farmer-level intervention can be estimated by pooling across MCC-level treatment arms, and conversely the effect of the MCC-level intervention can be estimated by pooling across farmer-level treatment arms. At the same time, Muralidharan, Romero, and Wüthrich (2023) cautions against indiscriminate pooling of treatment cells when the underlying interventions may exhibit interaction effects. The authors note that, in the presence of even modest complementarities or antagonisms between treatment components, pooling can lead to biased or misleading estimates of marginal treatment effects. Consequently, they argue that a more rigorous and defensible approach is to estimate fully interacted specifications, which allow each treatment combination to have its own effect and thereby avoid imposing potentially unwarranted additivity assumptions. One way to recover an unbiased estimate of the pooled main treatment effect is to consider the orthogonal treatment as a co-variate and adjust for it, entering it in the regression demeaned and fully interacted as in equations 22 and 23.[8]

[8] It is important to note that the estimand of the respective treatments in Equations 22 and 23 changes. For instance, while the estimand for the MCC-level treatment (T1) in Equation 21 is the effect of the treatment in and of itself, the estimand in Equation 22 is a weighted average of the pure effect of the MCC-level treatment (T1) and the effect of the T1 in the presence of T2.

$$y_{i,m} = \alpha + \beta_{H2} T1_{i,m} + \beta_{H3}^{T1} \left(T2_{i,m} - \bar{T2}\right) + \beta_{H4}^{T1} T1_{i,m} \left(T2_{i,m} - \bar{T2}\right) + \beta_b . y_{i,m}^b + \varepsilon_{i,m} \tag{22}$$

$$y_{i,m} = \alpha + \beta_{H3} T2_{i,m} + \beta_{H2}^{T2} \left(T1_{i,m} - \bar{T1}\right) + \beta_{H4}^{T2} T2_{i,m} \left(T1_{i,m} - \bar{T1}\right) + \beta_b . y_{i,m}^b + \varepsilon_{i,m} \tag{23}$$

Next, note that β_{H4}^* in equation 21 is estimated as an incremental effect to the main effects. In our analysis, we will also test Hypothesis 4 by directly comparing farmers that received the training and are connected to an MCC that received a milk analyzer to farmers that did not get the training and are connected to a control MCC. In other words, we will estimate:

$$y_{i,m} = \alpha + \beta_{H4} . T1_{i,m} . T2_{i,m} + \beta_b . y_{i,m}^b + \varepsilon_{i,m} \tag{24}$$

after dropping from our data farmers that were exposed to only a single treatment. This specification recovers the effect of the intervention as an integrated package that addresses quality upgrading through both enforcement (the analyzer) and knowledge (the information treatment) channels.

Finally, we will also look at treatment heterogeneity, exploiting the fact that we stratified farmers on their link to the MCC. In particular, $C_{i,m}$ is an indicator variable at the farmer level that is one if farmer i is directly connected to MCC m (and zero if the farmer is connected through an intermediary). We also add a full set of interactions with this connection indicator, resulting in the following model:

$$y_{i,m} = \alpha + \alpha_C C_{i,m} + \beta_{H2T} . T1_{i,m} + \beta_{H3T} T2_{i,m} + \beta_{H4T} T2_{i.m} . T1_{i,m}$$

$$+ \beta_{H2C} . T1_{i,m} . C_{i,m} + \beta_{H3C} T2_i . C_{i,m} + \beta_{H4C} T2_i . T1_{i,m} . C_{i,m} + \beta_b . y_{i,m}^b + \varepsilon_{i,m} \tag{25}$$

This specification is used to test Hypotheses 5-7 outlined in Section 6.

In all regressions (Equations21 to 25), we report CR2 cluster-robust standard errors(Imbens and Kolesár, 2016) clustered at the level of randomization (MCC catchment area level) for descriptive purposes. However, because this is a randomized experiment with a known assignment mechanism, our primary inference is based on *Fisher randomization inference* (RI). We construct the permutation distribution of each treatment effect estimate under the sharp null hypothesis of no effect for any unit, using 5,000 random reassignments of treatment status that respect the experimental design: T1 is permuted across MCCs (maintaining the number treated), T2 is permuted across farmers within each MCC. The RI p-value is the fraction of permutation test statistics at least as extreme as the observed statistic (two-sided). RI p-values are exact and require no distributional assumptions about errors or cluster structure, making them

particularly appropriate given that most catchment areas contain a single MCC, where asymptotic cluster-robust inference is unreliable.

To address multiple hypothesis testing within outcome families, we exploit a natural property of randomization inference. For each permutation under the sharp null, we compute the test statistic for every outcome in the family simultaneously and count how many cross the conventional significance threshold. The omnibus p-value is the fraction of permutations that produce at least as many individually significant results as we observe in the actual data. This directly tests whether the observed *pattern* of results---not any single coefficient---could have arisen by chance under the null of no treatment effect. For example, if we observe 3 out of 5 milk sample outcomes significant at the 5 percent level, the omnibus p-value is the probability of seeing 3 or more significant outcomes under the null. This approach eliminates the need for parametric multiple testing corrections (such as Bonferroni or Benjamini-Hochberg), as it controls the probability of falsely detecting a pattern of significance at any desired level. In addition, we combine primary outcomes into an Anderson (2008) summary index(Anderson, 2008); the RI p-value on this index provides a complementary omnibus test.

8 Sample and Timeline

Sample size was determined using a series of power simulations, details of which can be found in the pre-analysis plan. The primary outcome for these calculations was the price of milk, modeled at both the MCC and farmer level. At the MCC level, prices were assumed normally distributed with mean 1000 UGX/liter and standard deviation 50, while farmer-level prices were drawn with the MCC mean and a higher variance (SD = 100) to capture greater dispersion at that level.[9] We assumed the intervention would increase MCC prices by 30 UGX/liter (medium to large effect size), translate into a 40 UGX/liter increase for farmers (small to medium effect), and that the individual-level information treatment would generate a 25 UGX/liter increase (small effect), with a large interaction effect of 50 UGX/liter. Power was calculated for the joint test of all hypotheses being true at the 5 percent significance level using 1,000 simulations across varying combinations of clusters (100–130 MCCs) and farmers per cluster (10–40). Results showed that with about 125 MCCs and 20 farmers per MCC (total sample of about 2,500 farmers), power was just above 0.80 for a joint significance test, while power for individual hypotheses was substantially higher, ranging from 0.87 to 0.99. At the design stage, we decided to target 130 MCCs to account for attrition. We note that these power calculations were calibrated for a price-channel hypothesis; the enforcement model developed in Section 4 predicts null price effects. We discuss ex-post power for the outcomes that matter under the enforcement hypothesis in Section11.3.

[9] Processors set fairly stable procurement prices and MCCs adjust tightly around those. The nested nature of the sampling and variance parameters imply an intra-cluster correlation of approximately $\rho \approx 0.20$.

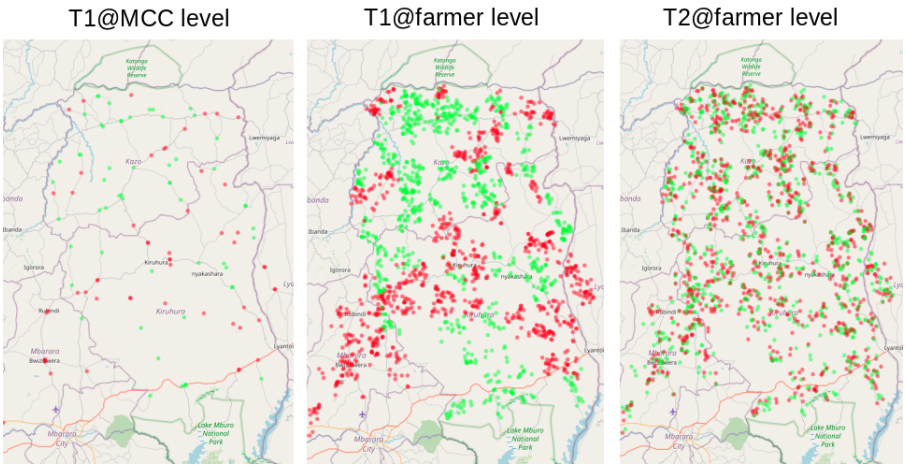

Figure 4: Sampling and Randomization

The field experiment was conducted in four districts of Southwestern Uganda: Ntungamo, Mbarara, Kazo, and Kirihura. The study began with a comprehensive census of all milk collection centers (MCCs) in the region, from which we sampled 130 MCCs. Half of these were randomly selected to receive T1, while the other half served as the control.[10] In each of the 130 MCCs, we then randomly selected 20 farmers who deliver milk to the center, half of which were taking the milk themselves and the other half using a trader or transported to get the milk to the MCC. Of these 20 farmers, 10 were randomly assigned to the video treatment (T2), while the remaining 10 served as the control group, with again in each group half directly connected and the other half connected through an intermediary.

Figure 4 illustrates the randomization of T1 and T2 in three of the four study districts at both the MCC and farmer levels. The leftmost panel displays the spatial distribution of MCCs and their assigned T1 status. The middle panel shows the corresponding T1 assignment for farmers connected to those MCCs, reflecting the fact that T1 was clustered at the MCC level. The rightmost panel depicts the allocation of T2 among farmers linked to MCCs, which was randomized at the individual level.

Baseline data from both MCCs and associated farmers was collected towards the end of 2022 using in-person surveys. At that time, the intervention at the farmer level (T2) was implemented. In the second half of 2023 milk analyzers were installed and MCC managers were trained on the use of the analyzers and the digital record-keeping system. Towards the end of the installation of the analyzers, in October 2023, we reinforced T2 by showing the video again

[10] In some cases, more than one MCC were in a particular location (eg. in a trading center). In such cases, MCCs were pooled in the same treatment group, as a result, the number of clusters was slightly lower than the number of MCCs.

to T2 farmers. Next, during the course of one year, together with the DDA we supported T1 MCCs with regular visits to make sure the equipment was operational, including recalibration after 6 months. In December 2024, endline data was collected. This involved in-person surveys at the MCC level and at the farmer level. In addition, during endline we also measured quality of incoming milk. To do so, an enumerator equipped with a milk analyzer paid an unannounced visit to each MCC enrolled in the study and measured all incoming milk samples over the course of one full day.[11]

9 Descriptive Statistics and Baseline Balance

We pre-registered 10 variables at each level to demonstrate balance. Results for five variables measured at the MCC level are in Table 2; five more variables are in Appendix Table A1 to conserve space. Table 2 shows that among MCCs in the control group, 63 percent are organized as cooperatives, reflecting how collective institutions continue to play a central role in the organization of Uganda's dairy value chain. Their average total storage capacity is about 4,000 liters, and 27 percent reported paying a premium for higher quality milk to suppliers. The average MCCs has been in operation for just under 10 years, and 58 percent facilitated the supply of acaracides to supplying farmer.

Appendix Table A1 further shows that among MCCs in the control group, the average number of full-time employees is about three, and a typical MCC receives milk from about 56 farmers or traders on an average day during the rainy season. On average, MCCs use 38 percent of their processing or cooling capacity during the dry season, which is indicative of significant seasonality affecting the sector. MCCs report they own 21 milk cans and the vast majority indicate that they provide credit or loans to cooperative members and regular suppliers.

To assess pre-treatment balance across treatment arms, both tables further show treatment–control differences for each baseline variable for T1 in the second column. Only one of the individual differences is statistically significant at the 5 percent level (total capacity of milk tanks, with treatment MCCs having higher capacity on average).[12] At the bottom of both tables we report results from an omnibus Wald test of joint balance across all covariates (F-statistic and p-values below). These tests do not reject the null of joint equality, indicating that treatment assignment is not systematically predicted by observed baseline

[11] This exercise posed logistical challenges, as MCCs opened early (usually at 8:00 am) and were often in remote areas. Enumerators frequently stayed overnight nearby to set up equipment before deliveries and carried generators for use where power was unavailable. To limit information spillovers, the testing was conducted over a short period with as many enumerators deployed simultaneously as possible.

[12] With 10 balance variables tested at the 5 percent level, one significant difference is consistent with chance alone. To verify that this imbalance does not drive our results, we note that our primary MCC-level outcomes are adoption and pricing variables rather than volume measures, and that ANCOVA specifications control for baseline values of the outcome where available. Tank capacity is not a direct determinant of the quality outcomes that constitute our main findings.

Table 2: Balance of Baseline Characteristics Between Treatment and Control Milk Collection Centers (MCCs)

	mean ctrl	analyzer	nobs
MCC is cooperative? (1=yes)	0.633	-0.086	124
	(0.486)	(0.09)	
Total Capacity of milk tanks (in liters)	4053.167	1031.115*	124
	(1809.592)	(433.343)	
MCC pays quality premium (1=yes)	0.267	-0.029	123
	(0.446)	(0.082)	
MCC age in years of operation	9.325	0.235	123
	(7.635)	(1.588)	
Facilitates supply of acaracides? (1=yes)	0.583	-0.068	124
	(0.497)	(0.093)	
F-statistic		1.93	
p-value		0.113	

Note: This table reports baseline balance between MCCs assigned to the analyzer treatment and those in the control group. Column 1 presents control group means with standard deviations in parentheses. Column 2 shows the difference in means (treatment minus control) with robust standard errors in parentheses. Column 3 lists the number of non-missing observations. "Total tank capacity" is measured in liters and "MCC age" in years. The F-statistic and p-value correspond to a joint test of equality of all covariates. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

characteristics. Overall, the tables suggest that randomization achieved balanced treatment and control groups along key baseline characteristics of MCCs for T1.

Table 3 reports baseline characteristics of dairy farmers across treatment groups. In the control group (corresponding to farmers that did not get the information on what quality parameters are important and are connected to an MCC that did not receive the milk testing and tracing bundle), household heads are on average 54 years old, with herds of about 62 animals, of which more than 90 percent are improved breeds.[13] Farmers sell roughly 60 liters of milk per day during the rainy season, and they spend on average USD 66 per month on chemical purchases—primarily acaracides. Further characteristics are in Appendix Table A2 and shows that the average household counts nearly ten members and produces about 70 liters of milk per day during the rainy season. Roughly 79 percent of farmers sell most of their milk to a collection center, three-quarters report using only steel containers when transacting, and a similarly high share are members of a dairy cooperative.

The farmer-level balance tables (Table 3 and Appendix Table A2) report six different sets of comparisons (corresponding to the six $\beta_{H.}$ coefficients in Equa-

[13]To calculate herd, we did not simply ask total numbers but asked 6 separate questions: we ask how many local cows; local heifers; and local calves the farmer has, and ask the same 3 categories for improved animals. Farmers were allowed to indicate that they did not know for a particular category, which were treated as missing in our analysis leading to the reduction in the number of observations.

Table 3: Baseline Balance Between Control Farmers and Farmers Assigned to Analyzer, Video, or Bundle Treatments

	mean ctrl	analyzer	video	bundle	nobs
Household Head Age (years)	54.469	-0.086	-0.584	-1.16	2261
	(12.633)	(1.148)	(1.123)	(1.503)	
Current Total herd size (number)	62.297	12.472	4.127	-9.378	1976
	(54.007)	(7.702)	(4.356)	(7.894)	
Number of improved animals in total herd (share)	0.932	-0.019	0.014	-0.002	1976
	(0.171)	(0.015)	(0.012)	(0.017)	
Liters milk sold per day (on average in the rainy season) (liters)	59.697	10.65	8.516^{+}	-13.539^{+}	2261
	(56.325)	(7.679)	(4.311)	(7.218)	
Average monthly expense (USD) on chemical purchases	65.551	33.097^{+}	-0.896	-34.363	904
	(91.1)	(18.869)	(12.42)	(22.723)	
F-statistic		0.268	0.497	1.241	
p-value		0.927	0.776	0.314	

Note: This table reports baseline balance across farmers in the control group and the three treatment groups (analyzer, video, and bundle). Column 1 shows control group means with standard deviations in parentheses. Columns 2–4 report differences in means relative to the control group, with robust standard errors in parentheses. Column 5 gives the number of non-missing observations for each variable. The F-statistic and p-value correspond to joint tests of equality of all covariates across groups. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

tion 25). First, we test for pre-treatment differences between farmers connected to MCCs that received a milk analyzer and those connected to control MCCs (corresponding to β_{H2T} and indicated as *analyzer* in the table heading). We find that farmers that are connected to a treatment MCC are somewhat less likely to be a member of a cooperative and more likely to be using steel cans or buckets. They also have a slightly higher expenditure on chemicals. However, F-tests do not reject joint orthogonality.

Second, we examine differences between farmers who received T2 and those who did not (corresponding to β_{H3T}; see column with header *video*). Again, some significant differences are found between the two groups (treated farmer produce and sell slightly more milk, and are more likely to use steel cans or buckets), but the joint tests are not significant.

Third, we report the interaction effects at baseline (denoted as *bundle* in the table and corresponding to β_{H4T}), where we now find some evidence that farmers that received the full package were producing and selling less milk, and were less likely to use steel cans and buckets. However, also here, joint significance tests do not suggest imbalance.

In the appendix, partly for consistency with results reporting below, we add a few more comparisons. In particular, in Table A3 we report on pre-treatment differences based on the pooled models (with the column named *analyzer* corresponding to β_{H2} in Equation 22 and the column named *video* corresponding to β_{H3} in Equation 23) for the 10 preregistered farmer level characteristics. In the last column, we test pre-treatment differences for the bundled intervention by directly comparing pure control farmers to those that were exposed to both T1 and T2, corresponding to β_{H4} in Equation 24. These additional models confirm good overall balance.

10 Attrition and Compliance

Meeting sample size targets within the available budget proved challenging at baseline, largely because dairy farmers are geographically dispersed. In the end, we surveyed 2,261 farmers across 124 MCCs, corresponding to 87 and 95 percent of our planned farmer and MCC samples, and 90 and 99 percent of the required sizes from the power calculations. Of these 124 MCCs, 60 were randomly assigned to control condition and 64 to treatment condition T1.

By endline, we managed to re-contact 122 MCCs, of which 2 MCCs refused to cooperate, leaving us with data for 120 MCCs. One of the 4 attrited MCCs was originally assigned to the control arm of the mcc level treatment; 3 were from the treatment group. A chi-square test confirms that attrition was unrelated to the treatment. We further managed to contact 2,139, from which 6 refused to be interviewed, leading to an effective sample of 2,133. Further restricting farmers to the subset of farmers connected to an MCC that we were able to interview at endline, we get 2,059. Also at the farmer level, attrition is unrelated to the treatments and any interaction thereof.

We also documented several compliance challenges. In some cases, analyzers

were moved to control MCCs, and in others, treatment MCCs did not use or retain the machines. At endline, we observed that 14 of the 59 control MCCs had a milk analyzer on site, and 14 of the 61 treatment MCCs did not have a machine readily available. Among the 14 analyzers found in control MCCs, five were devices originally provided through the project. In the treatment group, we identified 47 analyzers across MCCs, but only 37 were operational or could be used for immediate testing at the time of the visit; the remaining units were non-functional or lacked an available operator.

Similar compliance concerns arise at the farmer level. Although the information treatment was delivered in a controlled manner and the video was shown twice, only about half of farmers assigned to the video treatment recalled having been shown a video at endline, suggesting substantial recall decay over the study period. At the same time, more than one in five control-group farmers stated that they had seen a video explaining how to improve compositional quality, indicating potential contamination from social interactions or confusion with other extension materials. Among treated farmers, only 42 percent reported using the seed provided by the project, a rate that is plausibly affected by drought conditions during the season. We revisit these compliance concerns when discussing secondary outcomes in Section11.2 and when assessing threats to validity in Section11.3.

11 Results

11.1 Primary outcomes

11.1.1 MCC level

We pre-registered six primary outcomes at the level of the MCC. The enforcement model developed in Section4 generates clear predictions for these outcomes: MCCs should adopt and use the analyzers to enforce quality thresholds (Prediction 5), but prices and quality premia need not change under the flat-price regime (Prediction 3). To capture the aggregate effect across these outcomes (and simultaneously consider multiple hypothesis testing concerns), we also construct a summary index following Anderson (2008).

Our first two outcomes measure the extent to which quality testing becomes embedded in routine operations at the MCC. Specifically, we record whether MCCs used a milk analyzer to measure butter fat and solid-non-fat content of incoming milk samples in the last 7 days. In addition, we asked whether MCCs test outgoing milk deliveries destined for buyers. This measure is based on more detailed sales transaction data, and is true if for any buyer (eg processor, other trader, etc) butter fat or SNF was tested using a milk analyzer. Together, these indicators capture MCC uptake of the enforcement technology, the necessary precondition for the enforcement channel to operate.

The next two outcomes measure prices: the average price at which MCCs purchase milk from farmers during the seven days preceding the survey, and the average price MCCs received for sales to various buyers during the last transac-

Table 4: Primary Outcomes for Milk Collection Centers (MCCs)

	mean ctrl	analyzer	nobs
Tested incoming milk using MA (1=yes)	0.288	0.533**	120
	(0.457)	(0.081)	
Testing outgoing milk using MA (1=yes)	0.203	0.285**	120
	(0.406)	(0.09)	
Price at which milk was bought from farmers (UGX)	1075	-15.213	115
	(92.556)	(16.362)	
Price at which milk was sold (UGX)	1199.576	3.439	108
	(106.327)	(19.223)	
Does the MCC pay a quality premium to suppliers?	0.186	-0.032	119
	(0.393)	(0.066)	
Did the buyer pay a quality premium?	0.186	0.031	119
	(0.393)	(0.071)	
Index of primary MCC outcomes	-0.077	0.142	103
	(0.477)	(0.102)	

Note: Column 1 reports the mean of each outcome for the control group, with standard deviations in parentheses. Column 2 shows the estimated treatment effect of the analyzer intervention at the MCC level, with standard errors in parentheses. Column 3 lists the number of MCCs with non-missing observations. All prices are in Ugandan shillings (UGX). The "Index of primary MCC outcomes" is an Anderson (2008) index of the variables listed above. P-values are based on Fisher randomization inference (5,000 permutations); standard errors are CR2 cluster-robust at the catchment-area level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

tion in the previous week. The model predicts that these prices should remain unchanged under the existing flat-price institutional structure (Prediction 3).

Finally, two outcomes directly capture whether price differentiation by quality emerges. We ask whether the MCC pays explicit quality premiums to its suppliers, and whether downstream buyers pay a premium for higher-quality milk. Under the enforcement model, which operates through rejection risk at a fixed threshold rather than through quality-differentiated pricing, we do not expect these indicators to change.

Results are in Table 4. Consistent with Prediction 5, MCCs in the treatment group were substantially more likely to test incoming deliveries: the probability of testing rose by 53 percentage points relative to a control group mean of 29 percent, a large and highly significant increase. We also observe a large and significant increase in the probability of testing outgoing deliveries to buyers. These results confirm that MCCs adopted the analyzers as enforcement tools, raising the detection probabilityδ for incoming supply---the key parameter shift in the model.[14]

[14]These self-reported outcomes may raise concerns about experimenter bias. However, as shown in Table7, we obtain similar results when using direct enumerator observations of analyzer use.

Turning to prices, we find no evidence that the intervention affected transaction terms at MCCs. In the control group, the average farm-gate price paid to suppliers was about UGX 1,075 per liter, and we detect no significant increase in the treatment group. Similarly, the average price MCCs received from buyers was about UGX 2,200 per liter in the control group, with no evidence that the MCC-level treatment raised buyer offers. Critically, this null result is not a failure of the intervention but rather is precisely what the enforcement model predicts (Prediction 3): the analyzer operates through rejection risk at a fixed price, not through quality-differentiated pricing.

In addition, about 20 percent of MCCs in the control group reported paying a quality premium to suppliers, and the same share (11 out of 59) reported receiving a premium from buyers. The intervention did not increase the likelihood of such premiums being paid. This is consistent with Prediction 3: under the flat-price institutional regime, the enforcement channel raises quality without requiring any change in the price schedule.

In light of the compliance issues discussed in the previous section, we also estimated Local Average Treatment Effects (LATE), instrumenting the actual availability of a functioning milk analyzer with the randomized treatment assignment (See Appendix TableA4). The conclusions remain largely unchanged. Overall, the MCC-level results confirm the first link in the enforcement mechanism: the technology was adopted and used for screening (Prediction 5), while behavior changed without any accompanying shift in prices (Prediction 3). This pattern---testing up, prices flat---is the signature of the enforcement channel operating within a flat-price regime.

11.1.2 Farmer level

We also define four primary outcomes at the farmer level and combine them into an index to assess overall impact of the interventions on farmer-level outcomes. The enforcement model predicts modest increases in quality effort (Prediction 2), complementarity between the analyzer and the information treatment (Prediction 6), no change in prices (Prediction 3), and weaker responses among farmers selling through traders (Prediction 7).

The first outcome is a composite in itself, and measures production investment and management practices that are expected to improve milk quality. We construct an Anderson (2008) index from farmer reports on five recommended practices undertaken in the past year: (i) over-sowing new fields with improved forage grasses such as *Napier*, *Brachiaria*, or *Rhodes* grasses; (ii) using legume pastures such as *Centro*, *Siratro*, or *Desmodium*; (iii) adopting controlled or zero grazing during the last dry season; (iv) practicing pasture conservation through hay, silage, or haylage; and (v) using feed supplements such as maize bran, crop residues (for example banana peelings), or mineral licks. This index captures the intensive-margin quality effort e_i in the model: costly ongoing investments in feed and management whose returns are bounded by the flat-price regime.

The second outcome captures whether buyers actively checked milk quality at the point of transaction. Here, we distinguish between different categories of

buyers (eg. MCCs, processor, trader, etc.) and ask if milk was tested using a milk analyzer if the farmer sold milk to this particular buyer in the 7 days prior to the survey. The indicator is true if milk sold in the last 7 days was tested by at least one buyer.

The third outcome is the price received for milk sold. Specifically, we record the average price per liter obtained in the farmer's most recent sale during the past seven days, inclusive of any quality premiums. Farmers reported prices by buyer category, and we construct a weighted average across categories using transaction volumes as weights.[15] We also directly ask whether the buyer pays a higher price for higher-quality milk. Under the enforcement model, neither prices nor quality premia are expected to change (Prediction 3), since the enforcement channel operates through rejection risk at a fixed price rather than through quality-differentiated payment.

Table5 reports treatment effects on our four primary outcomes at the farmer level, as well as the index. The table shows parameter estimates from the fully interacted model of Equation 25 in Section 7. In the first column, we show mean outcome for pure control farmers (that did not get the information treatment and are connected to control MCCs); standard deviations are in parentheses below. In the second column, we provide an estimate of the farmer level main treatment effect of T1 (corresponding to β_{H2C} in Equation 25, standard errors in parentheses below) and in the third column, we provide an estimate of the farmer level main treatment effect of T2 (corresponding to β_{H3C} in Equation 25, standard errors in parentheses below). The fourth column provides an estimate of the interaction between the MCC level treatment and the farmer level treatment (corresponding to β_{H4C} in Equation 25, standard errors in parentheses below). Finally, below the standard errors, we also provide p-values for the null hypothesis that there is no difference in the estimated effect depending on how the farmer is connected to the MCC (directly or through a small trader). More in particular, the number in square brackets in column two is the p-value for the hypothesis $\beta_{H2C} \neq \beta_{H2T}$ obtained from equation 25. Similar tests are reported for $\beta_{H3C} \neq \beta_{H3T}$ in column three and $\beta_{H4C} \neq \beta_{H4T}$ in column four.

In addition to the tables in the main text, we again provide various tables with additional estimates in the Appendix for robustness. In particular, we present tables with results from the pooled regressions corresponding to Equations 22 and 23 and the treatment bundle corresponding to Equation 24 (See Appendix Tables A5 and A7). Furthermore, in light of the compliance issues reported in Section 10, we also present LATE estimates where in addition to

[15] Reported prices below 600 UGX/liter and above 1,500 UGX/liter are treated as implausible and set to missing. These thresholds correspond to roughly half and 1.4 times the control-group mean price, respectively, and affect fewer than 2 percent of observations. Similarly, self-reported daily volume quantities exceeding 500 liters are treated as implausible and excluded. To verify that our results are not sensitive to these specific cutoffs, we conduct a systematic sensitivity analysis varying both price bounds (500--1500, 600--1200, 600--1400, 700--1500, and no trimming) and volume caps (300, 500, 700, 1000 liters, and no cap). Results are stable across all threshold combinations. For volume outcomes that are right-skewed, we apply the inverse hyperbolic sine (IHS) transformation. Results in raw levels and log(x+1) are qualitatively identical.

Table 5: Primary Outcomes for Farmers

	mean	analyzer	video	bundle	nobs
Production investment and management (Index)	0.014	-0.068	-0.024^{+}	0.135*	2054
	(0.562)	(0.051)	(0.036)	(0.056)	
		[0.173]	[0.129]	[0.005]	
Buyer checked for quality (1=yes)	0.157	0.058^{+}	0.048	0.028	1337
	(0.364)	(0.057)	(0.039)	(0.057)	
		[0.436]	[0.082]	[0.882]	
Price received for milk sold	1019.292	13.674	5.297	-2.895	1254
	(99.211)	(14.375)	(11.359)	(15.037)	
		[0.691]	[0.828]	[0.855]	
Get quality premium	0.081	-0.001	0.042	-0.054	1281
	(0.274)	(0.036)	(0.035)	(0.045)	
		[0.753]	[0.127]	[0.176]	
Index of primary farmer outcomes	-0.036	0.092^{+}	0.087	0.013	1202
	(0.488)	(0.072)	(0.071)	(0.092)	
		[0.452]	[0.301]	[0.757]	

Note: Column 1 reports mean outcomes for pure control farmers (those not receiving the information treatment and connected to control MCCs), with standard deviations in parentheses. Columns 2–4 present the pooled main treatment effects of the MCC-level intervention (T1), the farmer-level information intervention (T2), and their combined exposure, respectively; robust standard errors appear in parentheses. Numbers in square brackets report p-values for tests of whether the treatment effect differs by the farmer's connection type (directly to an MCC versus through a small trader). Column 5 shows the number of non-missing observations. Prices are in Ugandan shillings (UGX). The "Index of primary farmer outcomes" is an Anderson (2008) index of the outcomes listed above; the RI p-value on this index is an exact omnibus test of the joint null. P-values are based on Fisher randomization inference (5,000 permutations); standard errors are CR2 cluster-robust at the catchment-area level. Significance levels: **, * and + denote significance at the 1, 5 and 10 percent level based on RI p-values.

instrumenting actual presence of a milk analyzer with MCC level treatment assignment, we now also instrument the fact that farmers recall having seen the video with the farmer level treatment assignment (See Appendix Tables A6 and A7).

Looking at the main (fully interacted) specification in Table5 and the pooled estimates in TableA5, we do not find evidence that the MCC-level intervention alone affected farmers' production investments and management practices. The farmer-level information treatment also shows no significant effect in the fully interacted model. However, the pooled regression suggests a modest positive effect of the farmer-level intervention (significant at the 10 percent level). This modest magnitude is consistent with Prediction 2: because the return to quality effort comes entirely from reducing the probability of rejection near the threshold---not from earning a price premium---the incentive to invest is bounded, and the model predicts only a limited behavioral response on the intensive margin.

The significant interaction term in Table5 indicates that this effect is driven almost entirely by farmers whose MCCs were also equipped with an analyzer. This is direct evidence for Prediction 6 (complementarity): the information intervention raises the productivity of effortθ_i, but this only translates into behavioral change when detection is also high, because without the threat of rejection there is no return to improving quality. Conversely, information alone---without enforcement---has limited effect, as the model predicts. We also find significant heterogeneity with respect to intermediation for the interaction effect, consistent with Prediction 7: the significant increase in investment and effort is driven by directly connected farmers, for whom the individual detection probability is higher than for those selling through traders who pool milk from multiple sources.

Among pure control farmers, approximately 16 percent report that their milk was tested using an analyzer. Farmers linked to treatment MCCs report slightly higher testing rates---by roughly 6 to 7 percentage points in both the fully interacted and pooled specifications---although these differences are not statistically distinguishable from zero.[16] The coefficient on the farmer-level information treatment is likewise positive but imprecisely estimated. At the same time, the fully interacted model indicates heterogeneity by connection type: the information treatment reduces the likelihood of being tested among farmers who supply through traders, consistent with Prediction 7. We detect no statistically significant interaction between the MCC-level and farmer-level interventions for this outcome.

Across all specifications, we find no evidence that either intervention affected

[16] Although these effects are generally small and imprecise, they are consistent with qualitative reports from MCC managers who stated that analyzers were typically used only in suspected cases of adulteration rather than on every single delivery. Our outcome measure captures self-reported testing across all potential offtakers, not just MCCs, and therefore may not fully reflect this selective-use pattern. Even so, selective but credible testing can still shift behavior: the possibility of being tested may be sufficient to deter adulteration, a mechanism supported by the compositional improvements documented in the sample-based analysis below (Section11.1.3). This is precisely the detection-probability channel formalized in the model: even stochastic testing raisesδ enough to cross the deterrence threshold for many farmers.

the prices farmers receive for milk. Control farmers report an average price of UGX 1,019 per liter---a level consistent with MCC-level data once accounting for sales to alternative buyers such as traders or direct consumers, who typically pay slightly less. Prices received by farmers in both treatment arms are statistically indistinguishable from those of control farmers, and this holds for the interaction term as well as for pooled, fully interacted, and instrumented models. Similarly, the likelihood that a farmer reports receiving an explicit quality premium does not differ across groups. These null price results mirror the MCC-level findings and are fully consistent with Prediction 3: in a flat-price regime where the enforcement channel operates through rejection risk rather than quality-differentiated payment, improved detection raises quality without altering the price schedule.

The overall composite index of primary farmer outcomes is not statistically significant. This is consistent with the model's prediction that the intensive-margin response (quality effort) is bounded: the return to effort under flat-price enforcement is inherently limited compared with the large discrete response on the extensive margin (adulteration), which is captured by the objective milk sample data analyzed in the next subsection. Nonetheless, the pooled specification does provide some suggestive evidence that the MCC-level intervention modestly improves the combined index, and a direct comparison between pure control farmers and those exposed to the full bundle of interventions points to a small but positive shift in behavior (See Appendix TableA5).

Interpreting the null pre-registered primary outcomes. It is important to be transparent that the composite index of pre-registered primary farmer outcomes is not statistically significant. This index was designed under a price-channel hypothesis---it combines improved practices, buyer quality checks, average sales price, and quality bonus receipt. Under the enforcement model we develop in Section 4, two of these four components (price and quality bonus) are predicted to show no effect: the model's central prediction is that enforcement operates through rejection risk, not price differentiation. Including these two outcomes that are null by construction under the enforcement model mechanically dilutes the composite index: the Anderson (2008) weighting scheme assigns positive weight to price and bonus, and their near-zero treatment effects drag the index toward zero even when the remaining components (improved practices, buyer checks) respond positively. A version of the index that excludes price and quality bonus---retaining only the two outcomes for which the enforcement model predicts positive effects---is significant. We nevertheless report the pre-registered specification exactly as planned, consistent with best practice for the use of pre-analysis plans(Duflo et al., Working Paper). The objective milk sample data analyzed below---which directly measures the behavior the enforcement model targets (adulteration and compositional quality)---provides the strongest test of the mechanism. The null composite index and the significant milk sample results are jointly consistent with the enforcement model: detection risk changes behavior on the extensive margin (stopping water addition) more than on the intensive margin (investing in quality practices), and it does not change prices.

11.1.3 Milk samples

The objective milk sample data provide the most direct test of the enforcement model's central predictions. Enumerators conducted supervised testing at both treatment and control MCCs. For one full day at each site, all incoming milk was analyzed using milk analyzers, yielding precise measures of key quality parameters: added water, butterfat, solids-not-fat (SNF), protein, and density (corrected lactometer reading). These variables are also combined into an index of overall quality.[17] The model generates sharp predictions for these data: a large reduction in added water (Prediction 1), a modest increase in compositional quality (Prediction 2), and a pronounced asymmetry between the two margins (Prediction 4).

It is important to note that the compositional quality measures obtained through supervised testing at MCCs are farmer-level outcomes, but they need not correspond to the same individuals who appear in the farmer survey. The universe of suppliers observed in the testing data reflects the set of farmers who delivered milk on the specific day of supervised measurement, which may include new entrants who began supplying after analyzers were introduced, as well as exclude farmers who had previously supplied but stopped delivering. Conversely, the survey sample includes baseline suppliers regardless of whether they continued or ceased deliveries by endline. Each data source therefore captures a different margin of adjustment. The MCC testing data reveal changes in the composition and quality of actual incoming supply, incorporating both behavioral responses and selection effects induced by the technology. In terms of the model, these data capture the combined effect of deterrence (farmers who stop adulterating) and selection (low-quality suppliers who stop delivering), both of which are predicted consequences of raising the detection probability. The survey-based outcomes, by contrast, capture how the intervention affected the behaviors and perceptions of a fixed cohort of farmers. Both perspectives are essential for identifying the mechanisms at play.

Table 6 reports the results obtained from the analysis of supervised testing data. The headline finding is a large and significant reduction in added water: milk delivered to treated MCCs contains nearly half a percentage point less added water relative to the control mean of 1.63 percent. This result provides strong support for Prediction 1---the extensive-margin deterrence effect---and is consistent with evidence that detection threats operate primarily on the extensive margin(Bott et al., 2020). In the model, even a moderate increase in the detection probabilityδ can push farmers past the critical threshold$\hat{\delta}_w$ at which adulteration is no longer profitable, producing a large discrete reduction in water addition. Density (CLR) rises by 0.44 units, consistent with lower dilution and greater compositional integrity.

Alongside this large adulteration reduction, we observe a modest increase in

[17]Samples collected during the initial days of fieldwork in Ntungamo district (before December 5, 2024) and in Kazo and Mbarara districts (before December 4, 2024) were excluded because a calibration error in the project Lactoscan was identified and corrected on those dates. Results are robust to including these observations.

Table 6: Milk quality

	mean ctrl	analyzer	nobs
Butter fat (%)	3.881	0.114**	2518
	(0.539)	(0.036)	
SNF (%)	8.579	0.081	2518
	(0.49)	(0.05)	
Added Water (%)	1.626	-0.488*	2518
	(3.682)	(0.213)	
Protein (%)	3.163	0.029	2518
	(0.194)	(0.022)	
Density (CLR)	27.93	0.435*	2518
	(2.31)	(0.197)	
Index	-0.103	0.209**	2518
	(0.817)	(0.054)	

Note: Column 1 reports mean milk quality measures for the control group, with standard deviations in parentheses. Column 2 presents the average treatment effect of the MCC-level analyzer intervention, with standard errors in parentheses. Column 3 shows the number of non-missing observations. The index is an Anderson (2008) index of the five quality components, with added water entering with a negative sign. P-values are based on Fisher randomization inference (5,000 permutations); standard errors are CR2 cluster-robust at the catchment-area level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

butterfat: average butterfat content rises from 3.88 percent in control MCCs to approximately 4.00 percent in treated MCCs. This is consistent with Prediction 2---the bounded intensive-margin response. The return to quality effort under a flat-price regime comes solely from reducing the probability of rejection near the acceptance threshold, and once a farmer's expected quality comfortably exceeds that threshold the incentive to invest further is weak. SNF shows a similar directional increase, but the estimate is not statistically distinguishable from zero with the current sample size. Protein---which is a component of SNF---shows no detectable change. These biologically determined components depend on breed and long-term feeding practices that are unlikely to respond to a short-run increase in detection probability.

The composite index combining all five components indicates a significant improvement of 0.21 standard deviations in overall milk quality. Crucially, this improvement is driven primarily by the large reduction in adulteration rather than by quality enhancement, exactly as the model predicts (Prediction 4). The pronounced asymmetry between the extensive margin (added water down sharply) and the intensive margin (fat up modestly, SNF and protein unchanged) reflects the fundamental cost asymmetry in the model: stopping adulteration requires only that the farmer cease an essentially costless action whose benefit is now outweighed by detection risk, whereas increasing compositional quality demands costly ongoing investment in feed and management with returns that

are bounded by the flat-price regime. Moreover, because the supervised testing data capture all actual deliveries on the day of measurement, the improvement reflects both behavioral change (deterrence among continuing suppliers) and selection (low-quality suppliers who ceased delivering once the detection probability rose). Both channels are predicted consequences of raisingδ.

To assess whether this pattern of results could have arisen by chance under the null of no treatment effect, we conduct an omnibus count test using the randomization inference framework. Three of the five individual milk sample outcomes are statistically significant at the 5 percent level. Under 5,000 permutations of the treatment assignment, the probability of observing three or more individually significant outcomes is 0.022. This omnibus p-value directly addresses the multiple comparisons concern without requiring parametric corrections: the observed pattern of significance across milk quality measures is unlikely to have been generated by chance. At the MCC level, two of six primary outcomes (incoming and outgoing testing rates) are individually significant; the omnibus p-value for observing two or more significant MCC outcomes under the null is 0.039.

The treatment-on-the-treated estimates, presented in Appendix TableA8 for completeness, are generally less precisely estimated than the intention-to-treat (ITT) effects. The instrument only shifts the probability that an MCC consistently uses the analyzer, and the first stage is mechanically weaker than the underlying ITT assignment. As a result, the 2SLS estimator inflates sampling variability because it attributes all treatment effects to the subset of MCCs that actually complied with analyzer use. Moreover, if assignment affects outcomes partly through channels other than actual machine use---such as expectations, monitoring incentives, or changes in manager behavior---then the instrument captures only a fraction of the relevant variation, further reducing precision. Together, these factors imply wider confidence intervals for the LATE relative to the more precise ITT estimates.

11.2 Secondary outcomes

11.2.1 MCC level

A key question in understanding the effects of the MCC-level intervention is whether the treatment actually changed the visibility, availability, and operational use of the milk analyzers. These intermediate outcomes form the core impact pathway through which the intervention could influence milk quality, rejection practices, and price formation. Table 7 therefore reports several secondary measures intended to capture the uptake and functioning of the technology as observed by enumerators during the endline visit. Because these outcomes rely on direct observation and on-the-spot demonstrations, they offer an objective assessment of whether the treatment altered day-to-day practices at MCCs.

The first outcome captures whether the enumerator observed the project posters publicizing the milk analyzers and encouraging farmers to request their

milk to be tested. This variable reflects both the visibility of the intervention to suppliers and the extent to which MCCs engaged in basic communication around quality measurement. The second outcome indicates whether a milk analyzer was physically observed at the MCC during the visit. A related variable distinguishes analyzers distributed through the project from machines obtained through alternative channels, thereby providing a direct measure of compliance with treatment assignment.

Functionality is assessed through an indicator based on the enumerator witnessing a successful real-time demonstration of the machine. This corresponds to the manager performing an on-the-spot test and the machine producing a result without technical failure. Finally, a last outcome variable captures whether the MCC used the digital record-keeping in the form of an application (including but not limited to the one developed as part of the project) or a file on a computer (generally an excel sheet).

The results show clear and sizable treatment effects on the visibility and presence of the technology. Treated MCCs are significantly more likely to display the poster, suggesting that assignment increased the outward communication of quality testing. The presence of a milk analyzer increases by roughly 53 percentage points, and the increase is even larger when restricting attention to analyzers supplied directly through the project. These effects confirm both compliance and the persistence of the physical hardware at the treatment sites.

The intervention also increased the likelihood that MCCs used digital tools to record quality and quantity traded. This reinforces the idea that the project induced complementary behavioral changes rather than simple adoption of a single physical asset. Because digital record-keeping addresses traceability and transparency---capabilities that strengthen the enforcement of quality thresholds at the accept/reject margin---this result helps explain how the enforcement channel operates in practice, producing measurable improvements in milk composition and reductions in adulteration even without price differentiation.

11.2.2 Farmer level:

To understand whether the farmer-level components of the intervention were implemented as intended, and to assess the earliest steps along the behavioral impact pathway, we also examine a set of secondary outcomes capturing farmers' treatment exposure, compliance, and basic conceptual understanding of compositional quality. Table 8 reports these outcomes.

A first outcome simply asks whether the farmer recalls having been shown the informational video on milk quality during baseline. This directly captures exposure with the information treatment. A second outcome records whether the farmer recalls the distribution of the pasture seed pack, which served as a simple extension-style nudge toward improved feeding practices. We also have an indicator variable for farmers actually planting the seed. This allows us to study exposure, recall, and behavioral follow-through.

The final outcome tests whether farmers can correctly identify practices that increase the compositional quality of milk. Farmers were presented with

Table 7: Uptake and Operational Use of Milk Analyzers at MCCs

	mean ctrl	analyzer	nobs
Poster is visible (1=yes)	0.034	0.343**	120
	(0.183)	(0.076)	
Milk Analyzer present (1=yes)	0.237	0.533**	120
	(0.429)	(0.081)	
Project Milk Analyzer is present (1=yes)	0.085	0.637**	120
	(0.281)	(0.076)	
Milk analyzer operational (1=yes)	0.237	0.369**	120
	(0.429)	(0.086)	
MCC uses digital record-keeping (1=yes)	0.237	0.369**	120
	(0.429)	(0.087)	
Index of uptake	-0.436	0.858**	120
	(0.44)	(0.118)	

Note: Column 1 reports the mean of each outcome for the control group, with standard deviations in parentheses. Column 2 shows the estimated treatment effect of the analyzer intervention at the MCC level, with standard errors in parentheses. Column 3 lists the number of MCCs with non-missing observations. The "Index of uptake" is an Anderson (2008) index of the variables listed above. P-values are based on Fisher randomization inference (5,000 permutations); standard errors are CR2 cluster-robust at the catchment-area level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

three pairs of options—each contrasting a hygiene-focused practice (such as washing hands or using clean containers) with a compositional-quality-focused practice (such as controlled grazing or using feed supplements). The individual answers are aggregated again using and Anderson (2008) index. This measure provides an objective assessment of whether the interventions shifted farmers' understanding of what determines higher-quality milk.

The MCC-level intervention generates no detectable effects on any of the farmer-level outcomes. It does not increase recall of the video or seed, nor does it increase seed usage or improve farmers' understanding of compositional quality. This is consistent with the intervention design: the MCC treatment targeted technology adoption and testing behavior at collection centers, not farmer education. The results therefore suggest limited spillover from improved analyzer use at MCCs toward farmer knowledge or practice, reinforcing the need for direct engagement when targeting farmer behavior.

In contrast, the video treatment produces clear and statistically precise effects on all but one outcome. Farmers assigned to the video intervention are significantly more likely to recall both the video and the seed distribution. The video treatment also raises the probability of using the distributed seed, though not universally so, likely due to the fact that the season was exceptionally dry and some farmers decided planting the seed would be a waste.

We do not find that the farmer level intervention increased knowledge in the

fully interacted model in Table 8. Results from the pooled model, reported in Table A9, are somewhat more encouraging. If we remove from the index the multiple choice questions we deem to be the most difficult, the coefficient does become significantly positive.

The composite index, which aggregates all farmer-level outcomes into a single measure, reinforces this conclusion. The video treatment leads to a substantial and statistically significant increase in overall uptake, while the MCC treatment shows no effect. Taken together, these results show that farmer-level behavioral change depends critically on direct information provision, whereas changes in practices at MCCs do not automatically trickle down to farmers without explicit communication or incentives.

11.3 Falsifiability and alternative explanations

A natural concern is whether the enforcement model is falsifiable---that is, what pattern of results would be inconsistent with it? We identify four such patterns. First, if added water had not decreased at treated MCCs, the model's central prediction (Prediction 1) would be rejected: the enforcement channel requires that adulteration falls when detection improves. Second, if prices had increased significantly at treated MCCs, this would favor a price-incentive channel over the enforcement channel, since Prediction 3 specifically states that prices should remain flat under the institutional regime we study. Third, if directly connected farmers and trader-connected farmers had responded identically, this would contradict the detection-dilution mechanism (Prediction 7). Fourth, if the information intervention had worked independently of the analyzer---rather than only in combination---this would undermine the complementarity logic (Prediction 6) and suggest that knowledge alone is sufficient to change behavior without enforcement.

Power considerations. Our original power calculations were calibrated to detect a 30 UGX/liter price effect at the MCC level under a price-channel hypothesis. Given that the enforcement model predicts null price effects, this raises the question of whether the study was adequately powered for the outcomes that matter under the enforcement hypothesis. For the milk sample data---where we observe the largest effects---the control-group standard deviation of percent added water is approximately 3.7 percentage points. With 92 clusters and over 2,500 milk samples, we have adequate power to detect effects of approximately 1 percentage point at conventional significance levels. The observed reduction in added water (0.49 percentage points, approximately 30 percent of the control mean) is well within the detectable range. For farmer-level survey outcomes, power is lower for interaction terms in the fully interacted model: with 92 clusters and 8 coefficients, the effective degrees of freedom are limited, and we interpret null interaction effects cautiously.

Selection versus deterrence. The milk sample results capture whoever delivered milk on the supervised testing day, not the fixed panel from the baseline survey. The reduction in added water could therefore reflect either behavioral change (continuing suppliers improving quality) or selection (low-quality sup-

Table 8: Secondary outcomes at farmer level - uptake

	mean	analyzer	video	bundle	nobs
Remembers video	0.209	-0.022	0.3**	0.012	2059
	(0.407)	(0.04)	(0.04)	(0.056)	
		[0.367]	[0.005]	[0.538]	
Remembers receiving seed	0.251	-0.002	0.32**	0.056	2059
	(0.434)	(0.046)	(0.043)	(0.054)	
		[0.929]	[0.038]	[0.26]	
Used seed	0.171	-0.01	0.213**	0.016	2059
	(0.377)	(0.043)	(0.043)	(0.056)	
		[0.632]	[0.22]	[0.485]	
Knows compositional quality matters	-0.037	0.064	0.029	-0.027	2059
	(0.767)	(0.086)	(0.055)	(0.087)	
		[0.285]	[0.915]	[0.373]	
Index of uptake	-0.211	0.017	0.345**	0.011	2059
	(0.571)	(0.057)	(0.05)	(0.07)	
		[0.813]	[0.031]	[0.888]	

Note: First column reports control group means (and standard deviations below); the second column shows average treatment effect of MCC level intervention, (standard error of coefficient estimate below) and [p-value for the hypothesis of no treatment heterogeneity with respect to connection type below]; the third column shows average treatment effect of the farmer level intervention, (standard error of coefficient estimate below) and [p-value for the hypothesis of no treatment heterogeneity with respect to connection type below]; the fourth column shows average treatment effect of the interaction of the MCC level and farmer level intervention, (standard error of coefficient estimate below) and [p-value for the hypothesis of no treatment heterogeneity with respect to connection type below]; P-values are based on Fisher randomization inference (5,000 permutations); standard errors are CR2 cluster-robust at the catchment area level. **, * and + denote significance at the 1, 5 and 10 percent level.

pliers ceasing to deliver). Both mechanisms are consistent with the enforcement model---the model predicts that some farmers below the critical threshold$\hat{\delta}_w$ will stop adulterating while others may exit---but the welfare implications differ. Our secondary outcomes on farmer switching provide indirect evidence: we find no significant difference in the probability that treated farmers remain connected to their MCC, suggesting that the quality improvement is not primarily driven by supplier exit. However, we cannot definitively decompose the effect because the milk sample and survey populations differ.

Compliance and the exclusion restriction. As documented in Section10, non-compliance was substantial in both directions: 14 of 59 control MCCs had a milk analyzer at endline, and 14 of 61 treatment MCCs did not have a functioning machine. This two-way non-compliance motivates our LATE estimates, which instrument actual analyzer availability with random assignment. However, the standard LATE interpretation requires the exclusion restriction---that random assignment affects outcomes only through the endogenous variable (analyzer use). Because the T1 treatment was a bundle (analyzer, tablet, application, posters, DDA support visits), assignment could plausibly affect outcomes through channels other than analyzer use per se. For instance, the ongoing DDA visits or the record-keeping infrastructure could independently change MCC behavior. We therefore urge caution in interpreting the LATE as the causal effect of analyzer use alone. The ITT estimates---which capture the joint effect of the full package as randomly assigned---remain our preferred specification and are not subject to the exclusion restriction concern. The LATE estimates are presented for completeness and to provide a sense of magnitude, but we acknowledge that they likely overstate the effect of the analyzer in isolation because they attribute the full bundle effect to a single compliance margin.

Video treatment contamination. A second challenge concerns the farmer-level information treatment. At endline, only 54 percent of farmers assigned to the video treatment recalled having been shown a video, suggesting either recall decay or imperfect implementation. More concerning, more than one in five control-group farmers reported having seen a video about compositional quality, indicating potential contamination. This contamination could arise from social interactions within villages, shared viewing of agricultural extension materials, or recall confusion with other media exposure. The consequence of such contamination is straightforward: ITT estimates for the video treatment are attenuated toward zero because some control farmers received (or believe they received) the treatment. This makes the video arm a conservative test of information provision. The significant effects we do find on recall and seed usage (Table8) should be interpreted as lower bounds. For the interaction between the analyzer and video treatments---where we find the strongest farmer-level effects---contamination in the video arm would similarly attenuate the estimated interaction, making our finding of complementarity conservative. We note that the main results of the paper concern the MCC-level analyzer treatment and the objective milk sample data, neither of which is affected by video contamination.

Experimenter demand and Hawthorne effects. The implementation of T1 in-

volved ongoing engagement with treatment MCCs: DDA technicians conducted periodic monitoring visits to ensure equipment functionality, and a WhatsApp group was maintained for troubleshooting and coordination. Control MCCs did not receive equivalent attention. This differential contact raises the possibility that behavioral changes at treatment MCCs partly reflect Hawthorne effects---responses to being monitored---rather than the enforcement mechanism per se. Several features of the results help bound this concern. First, the strongest effects appear in the objective milk sample data, which were collected during a single supervised testing day at both treatment and control MCCs using independent Lactoscan devices---the testing protocol was identical at all sites. Hawthorne effects would need to operate through the farmers delivering milk, who had no contact with the research team during the testing day. Second, the pattern of results is highly specific: the intervention reduced adulteration and increased testing rates but did not affect prices, quality premia, or most farmer-level outcomes. A generic Hawthorne response would be expected to improve all measured outcomes, not selectively improve those predicted by the enforcement model. Third, the interaction between treatments (analyzer×video) follows the predicted complementarity pattern, which is difficult to generate through demand effects. Nevertheless, we cannot fully rule out that the additional attention from DDA visits contributed to the treatment effect, and we flag this as a limitation.

Seed confound in the video treatment. The farmer-level information treatment (T2) included not only the video and cartoon handout but also a 1 kg bag of improved pasture seeds (Rhodes grass). Because the seed was distributed together with the video, any effects attributed to the information treatment may partly reflect the material input. In particular, the interaction effect (analyzer×video) could capture complementarity between enforcement and seed provision rather than between enforcement and knowledge. We note two mitigating factors. First, only 42 percent of treated farmers reported actually planting the seed, likely because of drought conditions during the season---limiting the scope for the seed to drive effects through improved feeding. Second, the seed treatment alone (without the analyzer) shows no significant effects on milk quality or farmer practices. The interaction effect is driven by the combination of enforcement at the MCC and farmer-level engagement, but we cannot cleanly separate whether the active ingredient in T2 is knowledge, the seed, or their combination. This is a limitation of the multi-component treatment design.

Project devices at control MCCs. Five of the 14 milk analyzers found at control MCCs were devices originally distributed through the project. These machines were likely moved or sold by treatment MCCs, which represents a leakage of the physical treatment to the control group. This contamination attenuates the ITT estimate because some control MCCs gained access to the technology. The remaining 9 analyzers at control MCCs were acquired independently, reflecting autonomous adoption that would have occurred absent the intervention. Both forms of analyzer presence in the control group work against finding treatment effects, making our ITT estimates conservative. In the LATE

specification, these control MCCs with analyzers are appropriately reclassified as compliers, and the instrument remains valid provided that assignment affects outcomes only through analyzer use (subject to the exclusion restriction caveat discussed above).

12 Conclusion

This study provides experimental evidence that measurement technology can improve agricultural product quality through an enforcement channel---raising the probability that low-quality supply is detected and rejected---rather than through price incentives. In a field experiment spanning 120 milk collection centers and over 2,500 dairy farmers in southwestern Uganda, we introduced digital milk analyzers that sharply increased the precision with which MCCs could evaluate incoming milk. We developed a simple model in which a single parameter shift---the detection probabilityδ---generates predictions for both extensive-margin (adulteration) and intensive-margin (quality effort) responses, along with a clear prediction that prices should not change under the prevailing flat-price institutional regime.

The empirical results align closely with the model's predictions. MCCs adopted the analyzers and used them to screen incoming supply (Prediction 5), confirming the necessary precondition for the enforcement channel to operate. The most striking finding comes from objective milk sample data: added water fell sharply at treated MCCs (Prediction 1), while butterfat increased modestly (Prediction 2). The pronounced asymmetry between these two margins (Prediction 4) reflects the fundamental cost asymmetry in the model---stopping adulteration requires only that a farmer cease an essentially costless action whose benefit is now outweighed by detection risk, whereas increasing compositional quality demands costly ongoing investment with returns bounded by the flat-price regime. Self-reported farmer survey data corroborate these patterns: quality effort increased modestly, and only when the information treatment was combined with the analyzer (Prediction 6), confirming that enforcement and information are complements. Directly connected farmers responded more strongly than those selling through traders (Prediction 7), consistent with the model's prediction that intermediation dilutes individual detection.

Critically, the intervention did not affect prices at any stage of the chain---neither MCC buying prices, MCC selling prices, farmer prices, nor the likelihood of quality premia being paid. Under the enforcement model, this is not a failure but the expected outcome (Prediction 3). The enforcement channel operates through rejection risk at a fixed price and threshold, not through quality-differentiated payment. Quality improved because the threat of rejection became credible, not because superior quality was rewarded with a higher price.

The absence of quality premia also reflects deeper institutional features of this market. In many dairy cooperatives, including those operating in our study area, farmers are paid a uniform price per liter regardless of compositional

quality. These uniform-price policies are defended as preserving solidarity and minimizing conflict among members, but they also remove the possibility of transmitting individual-level incentives for quality upgrading. In principle, cooperatives could respond to downstream quality premia at the collective level, even without differentiating prices internally, provided they establish credible mechanisms for managing the common-pool quality problem(Winfree and McCluskey, 2005; Bonroy et al., 2019). In practice, the combination of cooperative governance norms, biweekly price-setting cycles that lock in prices for 14-day periods(Casaburi and Macchiavello, 2019; Swinnen and Vandeplas, 2010), and a concentrated processing sector with limited competitive pressure to differentiate on quality jointly sustain the flat-price equilibrium.

The central contribution of this paper is to show that enforcement through detection can substitute for price incentives in driving quality upgrading. The conventional view holds that measurement technology improves quality by enabling price differentiation: once quality becomes observable, markets for quality emerge, and price premia reward superior supply. Our results demonstrate an alternative mechanism. Even when institutional constraints prevent quality premia from emerging, measurement technology can still improve quality by making the accept/reject decision at the aggregation point more precise. The technology shifts the enforcement frontier, not the price frontier.

This finding has implications beyond the Ugandan dairy sector. Many agricultural value chains in developing countries share the same basic structure: quality is difficult to observe at the point of aggregation, institutional constraints prevent quality-differentiated pricing(Hansman et al., 2020), and adulteration is a persistent problem. In tea, coffee, grain, and oilseed supply chains, aggregators face similar challenges in verifying the quality of individual lots within pooled shipments. Our results suggest that deploying low-cost measurement technology at the aggregation point can improve quality through the enforcement channel alone, without requiring the more demanding institutional changes needed to implement quality-based payment schemes. The enforcement channel operates through a simple mechanism---raising the detection probability makes adulteration unprofitable---that can function within existing institutional structures.

That said, the enforcement channel has inherent limitations. It is most powerful on the extensive margin, where the response is discrete (stop adulterating or not), and less effective on the intensive margin, where costly ongoing investment in feed and management is required. Our results confirm this: the large reduction in adulteration is accompanied by only modest gains in compositional quality. A transition from enforcement-based quality assurance to price-based quality incentives would likely unlock further quality improvements on the intensive margin, but requires institutional changes---in cooperative governance, in processor pricing practices, and in market competition---that go beyond the provision of measurement technology. There are encouraging signs: a new large processor specializing in infant formula production has recently entered the market and begun approaching MCCs about supplying under a quality-based payment scheme. Increased competition of this kind, combined with broader deployment of testing capacity, may eventually create the conditions for a transition from

flat-price enforcement to quality-differentiated pricing.

13 Acknowledgments

During the preparation of this work the author(s) used OpenAI's chatGPT in order to obtain editorial suggestions to improve clarity and readability and explore alternative phrasing for technical terms. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article. We would like to thank all funders who supported this research through their contributions to the CGIAR Trust Fund: https://www.cgiar.org/funders/. In particular, funding from the CGIAR Rethinking Food Markets Initiative and the CGIAR Science program on Better Diets and Nutrition is greatly appreciated. We want to thank Leocardia Nabwire, Wilberfoce Walukano and Marc Charles Wanume for field support. This research received clearance form Makerere's School of Social Sciences Research Ethics Committee (MAKSSREC-10.2022.594/AR) as well as from IFPRI IRB (DSGD-22-1057). The research was also registered at the Ugandan National Commission for Science and Technology (SS1520ES).

Online Appendix

Table A1: Balance of Baseline Characteristics Between Treatment and Control Milk Collection Centers (MCCs)

	mean ctrl	analyzer	nobs
Number of people employed at MCC?	2.967	0.346	124
	(1.886)	(0.337)	
Nr farmers that supply to MCC	55.923	-6.573	112
	(64.037)	(10.032)	
Capacity use during dry season (%)	37.978	-4.564	119
	(20.991)	(3.718)	
Number of milk cans owned by MCC	21.05	0.716	124
	(52.963)	(7.877)	
Supplies credit/loans (1=yes)	0.833	0.057	124
	(0.376)	(0.07)	
F-statistic		1.604	
p-value		0.269	

Note: This table reports baseline balance between MCCs assigned to the analyzer treatment and those in the control group. Column 1 presents control group means with standard deviations in parentheses. Column 2 shows the difference in means (treatment minus control) with robust standard errors in parentheses. Column 3 lists the number of non-missing observations. "Total tank capacity" is measured in liters and "MCC age" in years. The F-statistic and p-value correspond to a joint test of equality of all covariates. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Baseline Balance Between Control Farmers and Farmers Assigned to Analyzer, Video, or Bundle Treatments

	mean ctrl	analyzer	video	bundle	nobs
Household Members (number)	9.754	-0.122	-0.004	0.654	2261
	(4.687)	(0.436)	(0.31)	(0.478)	
Liters Produced Total Per Day (average during rainy season) (liters)	70.053	11.774	9.381^{+}	-14.316^{+}	2261
	(61.477)	(8.554)	(4.91)	(7.845)	
Normally during the rainy season sells most of its milk to a milk collection center? (1=yes)	0.793	0.01	0.001	0.017	2261
	(0.405)	(0.039)	(0.018)	(0.033)	
Uses only steel can/bucket during sales transactions in the last 7 days before survey? (1=yes)	0.752	0.1*	0.049^{+}	-0.078^{+}	2261
	(0.432)	(0.049)	(0.028)	(0.045)	
Member of dairy cooperative? (1=yes)	0.757	-0.104*	0	0.003	2261
	(0.429)	(0.051)	(0.022)	(0.038)	
F-statistic		1.283	0.605	1.322	
p-value		0.288	0.696	0.273	

Note: This table reports baseline balance across farmers in the control group and the three treatment groups (analyzer, video, and bundle). Column 1 shows control group means with standard deviations in parentheses. Columns 2–4 report differences in means relative to the control group, with robust standard errors in parentheses. Column 5 gives the number of non-missing observations for each variable. The F-statistic and p-value correspond to joint tests of equality of all covariates across groups. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Baseline Balance Between Control Farmers and Farmers Assigned to Analyzer, Video, or Bundle Treatments (pooled)

	mean ctrl	analyzer	video	nobs	
Household Head Age (years)	54.469	-0.835	-0.592	-1.83	2261
	(12.633)	(0.762)	(0.516)	(1.23)	
Current Total herd size (number)	62.297	2.63	3.37	7.221	1976
	(54.007)	(5.641)	(2.802)	(8.236)	
Number of improved animals in total herd (share)	0.932	-0.014	0.002	-0.006	1976
	(0.171)	(0.011)	(0.007)	(0.015)	
Liters milk sold per day (on average in the rainy season) (liters)	59.697	1.938	3.317	5.627	2261
	(56.325)	(5.626)	(2.434)	(7.371)	
Average monthly expense (USD) on chemical purchases	65.551	-2.534	0.339	-2.161	904
	(91.1)	(9.446)	(6.92)	(13.732)	
Household Members (number)	9.754	-0.048	0.162	0.528	2261
	(4.687)	(0.298)	(0.178)	(0.446)	
Liters Produced Total Per Day (average during rainy season) (liters)	70.053	4.198	2.343	6.84	2261
	(61.477)	(6.653)	(3.05)	(8.011)	
Normally during the rainy season sells most of its milk to a milk collection center? (1=yes)	0.793	-0.01	0.019	0.029	2261
	(0.405)	(0.043)	(0.014)	(0.036)	
Uses only steel can/bucket during sales transactions in the last 7 days before survey? (1=yes)	0.752	0.023	0.006	0.071	2261
	(0.432)	(0.036)	(0.015)	(0.05)	
Member of dairy cooperative? (1=yes)	0.757	-0.089^{+}	-0.001	-0.1^{+}	2261
	(0.429)	(0.045)	(0.014)	(0.053)	

Note: The first column reports means for the pure control group, with standard deviations in parentheses. The remaining columns report differences in means for each treatment arm relative to its appropriate control group in the pooled sample, with robust standard errors in parentheses. Standard errors are clustered at the catchment-area level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Primary Outcomes for Milk Collection Centers (MCCs) – LATE

	mean ctrl	analyzer	nobs
Tested incoming milk using MA (1=yes)	0.288	1.445**	120
	(0.457)	(0.246)	
Testing outgoing milk using MA (1=yes)	0.203	0.775**	120
	(0.406)	(0.232)	
Price at which milk was bought from farmers (UGX)	1075	-42.06	115
	(92.556)	(47.086)	
Price at which milk was sold (UGX)	1199.576	9.892	108
	(106.327)	(55.393)	
Does the MCC pay a quality premium to suppliers?	0.186	-0.086	119
	(0.393)	(0.191)	
Did the buyer pay a quality premium?	0.186	0.084	119
	(0.393)	(0.195)	
Index of primary MCC outcomes	-0.077	0.429	103
	(0.477)	(0.276)	

Column 1 reports the mean of each outcome for the control group, with standard deviations in parentheses. Column 2 shows the estimated treatment effect of the analyzer intervention at the MCC level, with standard errors in parentheses. Column 3 lists the number of MCCs with non-missing observations. All prices are in Ugandan shillings (UGX). The "Index of primary MCC outcomes" is an Anderson (2008) index of the variables listed above. P-values are based on Fisher randomization inference (5,000 permutations); standard errors are CR2 cluster-robust at the catchment-area level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Primary outcomes for farmers (pooled)

	mean	analyzer	video	bundle	nobs
Production investment and management	0.014	-0.02	0.044^{+}	0.024	2054
	(0.562)	(0.039)	(0.024)	(0.048)	
Buyer checks for quality	0.157	0.07	0.016	0.086	1337
	(0.364)	(0.047)	(0.021)	(0.052)	
Price received for milk sold	1019.292	11.308	1.44	12.293	1254
	(99.211)	(9.852)	(5.104)	(11.954)	
Get quality premium	0.081	-0.016	-0.006	-0.022	1281
	(0.274)	(0.023)	(0.014)	(0.028)	
Index of primary farmer outcomes	-0.036	0.079^{+}	0.038	0.119*	1202
	(0.488)	(0.047)	(0.028)	(0.059)	

Note: Column 1 reports mean outcomes for pure control farmers (those not receiving the information treatment and linked to control MCCs), with standard deviations in parentheses. Columns 2–4 report pooled average treatment effects for the MCC-level intervention (T1), the farmer-level information intervention (T2), and their joint exposure, respectively. Each estimate is calculated relative to its appropriate control group in the pooled sample; robust standard errors are shown in parentheses. Square-bracketed values report p-values for tests of whether each treatment effect differs by connection type (directly to an MCC versus through a small trader). Column 5 gives the number of non-missing observations. Prices are in Ugandan shillings (UGX). The "Index of primary farmer outcomes" is an Anderson (2008) index of the outcomes listed. Standard errors are clustered at the catchment-area level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Primary outcomes for farmers — LATE

	mean	analyzer	video	bundle	nobs
Production investment and management (Index)	0.014	-0.204	-0.122	0.396*	2054
	(0.562)	(0.159)	(0.082)	(0.173)	
		[0.185]	[0.053]	[0.015]	
Buyer checked for quality (1=yes)	0.157	0.184	0.021	0.098	1337
	(0.364)	(0.18)	(0.081)	(0.19)	
		[0.402]	[0.573]	[0.752]	
Price received for milk sold	1019.292	45.984	7.568	-9.206	1254
	(99.211)	(49.426)	(24.249)	(52.071)	
		[0.608]	[0.982]	[0.886]	
Get quality premium	0.081	-0.002	0.082	-0.16	1281
	(0.274)	(0.113)	(0.072)	(0.157)	
		[0.798]	[0.168]	[0.246]	
Index of primary farmer outcomes	-0.036	0.311	0.085	0.014	1202
	(0.488)	(0.256)	(0.138)	(0.3)	
		[0.399]	[0.653]	[0.828]	

Note: Column 1 reports mean outcomes for pure control farmers (those not receiving the information treatment and connected to control MCCs), with standard deviations in parentheses. Columns 2–4 present the pooled main treatment effects of the MCC-level intervention (T1), the farmer-level information intervention (T2), and their combined exposure, respectively; robust standard errors appear in parentheses. Numbers in square brackets report p-values for tests of whether the treatment effect differs by the farmer's connection type (directly to an MCC versus through a small trader). Column 5 shows the number of non-missing observations. Prices are in Ugandan shillings (UGX). The "Index of primary farmer outcomes" is an Anderson (2008) index of the outcomes listed above. Standard errors are clustered at the catchment-area level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Primary outcomes for farmers (pooled LATE)

	mean	analyzer	video	bundle	nobs
Production investment and management	0.014	-0.057	0.043^{+}	0.071	2054
	(0.562)	(0.113)	(0.024)	(0.137)	
Buyer checks for quality	0.157	0.195	0.02	0.256	1337
	(0.364)	(0.131)	(0.022)	(0.158)	
Price received for milk sold	1019.292	32.99	1.725	36.705	1254
	(99.211)	(29.183)	(5.311)	(36.979)	
Get quality premium	0.081	-0.042	-0.007	-0.062	1281
	(0.274)	(0.063)	(0.014)	(0.083)	
Index of primary farmer outcomes	-0.036	0.22^{+}	0.039	0.338^{+}	1202
	(0.488)	(0.131)	(0.03)	(0.173)	

Note: Column 1 reports mean outcomes for pure control farmers (those not receiving the information treatment and linked to control MCCs), with standard deviations in parentheses. Columns 2–4 report pooled average treatment effects for the MCC-level intervention (T1), the farmer-level information intervention (T2), and their joint exposure, respectively. Each estimate is calculated relative to its appropriate control group in the pooled sample; robust standard errors are shown in parentheses. Square-bracketed values report p-values for tests of whether each treatment effect differs by connection type (directly to an MCC versus through a small trader). Column 5 gives the number of non-missing observations. Prices are in Ugandan shillings (UGX). The "Index of primary farmer outcomes" is an Anderson (2008) index of the outcomes listed. Standard errors are clustered at the catchment-area level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Milk quality - LATE

	mean ctrl	analyzer	nobs
Butter fat (%)	3.881	0.395^{+}	2491
	(0.539)	(0.204)	
SNF (%)	8.579	0.247	2491
	(0.49)	(0.171)	
Added Water (%)	1.626	-1.533	2491
	(3.682)	(1.045)	
Protein (%)	3.163	0.091	2491
	(0.194)	(0.072)	
Density (CLR)	27.93	1.289^{+}	2491
	(2.31)	(0.737)	
Index	-0.103	0.675^{*}	2491
	(0.817)	(0.204)	

Note: Column 1 reports mean milk quality measures for the control group, with standard deviations in parentheses. Column 2 presents the average treatment effect of the MCC-level analyzer intervention, with standard errors in parentheses. Column 3 shows the number of non-missing observations. The index is an Anderson (2008) index of the five quality components, with added water entering with a negative sign. P-values are based on Fisher randomization inference (5,000 permutations); standard errors are CR2 cluster-robust at the catchment-area level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Secondary outcomes at farmer level (pooled) - Uptake

	mean	analyzer	video	bundle	nobs
Remembers video	0.209	-0.006	0.326**	0.29**	2059
	(0.407)	(0.031)	(0.021)	(0.05)	
Remembers seed	0.251	0.003	0.391**	0.374**	2059
	(0.434)	(0.029)	(0.02)	(0.047)	
Used seed	0.171	-0.004	0.25**	0.22**	2059
	(0.377)	(0.028)	(0.019)	(0.052)	
Knows compositional quality matters	-0.037	0.039	0.057	0.065	2059
	(0.767)	(0.069)	(0.035)	(0.091)	
Index of uptake	-0.211	0.016	0.405**	0.372**	2059
	(0.571)	(0.041)	(0.028)	(0.066)	

Note: First column reports control group means (and standard deviations below); the second column shows average treatment effect of MCC level intervention, (standard error of coefficient estimate below); the third column shows average treatment effect of the farmer level intervention, (standard error of coefficient estimate below); P-values are based on Fisher randomization inference (5,000 permutations); standard errors are CR2 cluster-robust at the catchment area level. **, * and + denote significance at the 1, 5 and 10 percent level.